Contents

1 Deriving the Majorana Pfaffian from the Partition Function

Here, we derive the pfaffian formulation of the path integral. We roughly follow the derivation given in PFAFFIAN PAPER. Usually, after doing a Suzuki-Trotter approximation, which splits the exponential into $N_\tau = \beta/\Delta_\tau$ imaginary time slices and then performing a Hubbard-Stratanovich transformation of some sort for each factor, one ends up with a partition function of the form

$$Z \approx \int \mathcal{D}(\phi)\, e^{-S_B[\phi]} \mathrm{Tr} \left\{ \prod_{t=1}^{N_\tau} \exp\left(-\frac{1}{4} \gamma^T A_t \gamma \right) \right\} = \int \mathcal{D}(\phi)\, e^{-S_B[\phi]} W[\phi], \tag{1}$$

where $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, ..., \gamma_{2N})^{\mathrm{T}}$ is a $2N$-dimensional vector of Majorana operators obeying the Clifford algebra $\{\gamma_i, \gamma_j\} = 2\delta_{i,j}$. A_t are matrices, which in general depend on some auxiliary field ϕ. This formulation is valid for a continuous auxiliary field. In the case of a discrete HS-transformation one can simply replace the path integral with a sum over all possible field configurations. We notice that the main complexity of the derivation lies in finding an analytical expression for the weight $W[\phi]$.
It will turn out very convenient to calculate the trace in an enlarged Hilbert space. This is accomplished by replacing $\boldsymbol{\gamma} \to \boldsymbol{f}^\dagger + \boldsymbol{f}$, where $\boldsymbol{f}$ is the vector of Dirac fermion operators. These operators are simply a mathematical tool, and have nothing to do with any actual physical Dirac fermion. This substitution does not change the algebra since

$$\left\{ f_i^\dagger + f_i, f_j^\dagger + f_j \right\} = 2\delta_{ij}. \tag{2}$$

It does however change the dimension of the Hilbert space. Because each Majorana operator carries half a degree of freedom we started out with a Hilbert space dimension $d = 2^N$. Now we have $2N$ Dirac fermions, which span a Hilbert space of $d' = 2^{2N}$. So when we perform the trace in this larger Hilbert space we need to compensate the overcounting by a factor of $\frac{1}{2^N}$. We therefore find

$$\begin{aligned} W[\phi] &= \frac{1}{2^N} \mathrm{Tr} \left\{ \prod_{t=1}^{N_\tau} \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^T A_t \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) \right\} \\ &= \frac{1}{2^N} \sum_n \langle n | \prod_{t=1}^{N_\tau} \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^T A_t \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | n \rangle , \end{aligned} \tag{3}$$

where n runs over all possible states in the Fock basis. In each factor we then insert a completeness relation of coherent states:

$$\mathbb{I} = \int \mathcal{D} \left(\psi^t, \bar{\psi}^t \right) | \psi^t \rangle \langle \psi^t | \, e^{-\left(\bar{\psi}^t \right)^{\mathrm{T}} \psi^t}, \tag{4}$$

where $\psi^t = \left(\psi_1^t, \psi_2^t, \ldots, \psi_{2N}^t \right)^{\mathrm{T}}$ and $\mathcal{D} \left(\psi^t, \bar{\psi}^t \right) = \prod_{x=1}^{2N} \mathrm{d}\psi_x^t \mathrm{d}\bar{\psi}_x^t$. This leads to

$$\begin{aligned} W[\phi] = \frac{1}{2^N} \int \mathcal{D} \left(\psi, \bar{\psi} \right) \sum_n \langle n | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_1 \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^1 \rangle \langle \psi^1 | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_2 \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^2 \rangle \ldots \\ \ldots \langle \psi^{N_\tau - 1} | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_{N_\tau} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^{N_\tau} \rangle \langle \psi^{N_\tau} | n \rangle \exp \left(- \sum_{t=1}^{N_\tau} (\bar{\psi}^t)^{\mathrm{T}} \psi^t \right) \\ = \frac{1}{2^N} \int \mathcal{D} \left(\psi, \bar{\psi} \right) \langle -\psi^{N_\tau} | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_1 \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^1 \rangle \langle \psi^1 | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_2 \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^2 \rangle \ldots \\ \ldots \langle \psi^{N_\tau - 1} | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_{N_\tau} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^{N_\tau} \rangle \exp \left(-\bar{\boldsymbol{\psi}}^{\mathrm{T}} \boldsymbol{\psi} \right) , \end{aligned} \tag{5}$$

where we have used the completeness relation of Fock states $\mathbb{I} = \sum_n |n\rangle \langle n|$ and the identity $\langle n | \psi \rangle \langle \psi | n \rangle = \langle -\psi | n \rangle \langle n | \psi \rangle$. We also introduced the vector $\boldsymbol{\psi} = \left(\psi^1, \psi^2, \ldots, \psi^{N_\tau} \right)^{\mathrm{T}}$. Because the entries of this vector have $2N$ components this vector ends up being $2NN_\tau$-dimensional. Now we need to deal with the matrix elements $\langle \psi^{t-1} | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_t \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^t \rangle$. Because the operators $\left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)_i$ obey the Clifford algebra every expression involving these operators may be brought into canonical form. This means that in every term of the expansion, each $\left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)_i$ can at most appear once and the operators appear in a set sequence (e.g. each operator string only has rising indices from left to right). In the canonical form each fermion operator f_i ($f_i^\dagger$) can act on the coherent state on the right (left) and the resulting expression can be written as a function of $\bar{\psi}^{t-1} + \psi^t$:

$$\langle \psi^{t-1} | \exp \left(-\frac{1}{4} \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right)^{\mathrm{T}} A_t \left(\boldsymbol{f}^\dagger + \boldsymbol{f} \right) \right) | \psi^t \rangle = \langle \psi^{t-1} | \, \omega \left(A_t, \bar{\psi}^{t-1} + \psi^t \right) | \psi^t \rangle . \tag{6}$$

Substituting this leads to

$$W\left[\phi\right]=\frac{1}{2^N}\int\mathcal{D}\left(\psi,\bar{\psi}\right)\omega\left(A_1,-\bar{\psi}^{N_\tau}+\psi^1\right)\omega\left(A_2,\bar{\psi}^1+\psi^2\right)\omega\left(A_3,\bar{\psi}^2+\psi^3\right)...\omega\left(A_{N_\tau},\bar{\psi}^{N_\tau-1}+\psi^{N_\tau}\right)$$
$$\exp\left(-\bar{\psi}^{N_\tau}\psi^1+\bar{\psi}^1\psi^2+\bar{\psi}^2\psi^3+...+\bar{\psi}^{N_\tau-1}\psi^{N_\tau}\right)\exp\left(-\bar{\boldsymbol{\psi}}\boldsymbol{\psi}\right)$$
$$=\frac{1}{2^N}\int\mathcal{D}\left(\psi,\bar{\psi}\right)\omega\left(A_1,-\bar{\psi}^{N_\tau}+\psi^1\right)\omega\left(A_2,\bar{\psi}^1+\psi^2\right)\omega\left(A_3,\bar{\psi}^2+\psi^3\right)...\omega\left(A_{N_\tau},\bar{\psi}^{N_\tau-1}+\psi^{N_\tau}\right)\exp\left(-\bar{\boldsymbol{\psi}}\boldsymbol{M}\boldsymbol{\psi}\right), \tag{7}$$

where we used $\langle\theta|\phi\rangle=\exp\left(\bar{\theta}^{\mathrm{T}}\phi\right)$ and introduced the matrix

$$\boldsymbol{M}=\left.\begin{bmatrix}\mathbb{I}_{2N} & -\mathbb{I}_{2N} & 0 & \cdots & 0\\ 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \ddots & \vdots\\ \vdots & \ddots & \ddots & \ddots & 0\\ 0 & \cdots & 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N}\\ \mathbb{I}_{2N} & \cdots & \cdots & 0 & \mathbb{I}_{2N}\end{bmatrix}\right\} 2NN_\tau\ . \tag{8}$$

By using the anticommutation relations we can rewrite this as

$$W\left[\phi\right]=\frac{1}{2^N}\int\mathcal{D}\left(\psi,\bar{\psi}\right)\ \omega\left(A_1,-\bar{\psi}^{N_\tau}+\psi^1\right)\omega\left(A_2,\bar{\psi}^1+\psi^2\right)\omega\left(A_3,\bar{\psi}^2+\psi^3\right)\cdots\omega\left(A_{N_\tau},\bar{\psi}^{N_\tau-1}+\psi^{N_\tau}\right)$$
$$\times\exp\left(-\frac{1}{2}\begin{bmatrix}\boldsymbol{\psi}^{\mathrm{T}} & \bar{\boldsymbol{\psi}}^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}0 & -\boldsymbol{M}^{\mathrm{T}}\\ \boldsymbol{M} & 0\end{bmatrix}\begin{bmatrix}\boldsymbol{\psi}\\ \bar{\boldsymbol{\psi}}\end{bmatrix}\right). \tag{9}$$

Because we initially enlarged the Hilbert space, we have doubled the degrees of freedom. We now want to integrate out all the uninvolved degrees of freedom. This is accomplished by a change of variables:

$$\begin{aligned}\theta^1&=\psi^1-\bar{\psi}^{N_\tau} & \tilde{\theta}^1&=\psi^1+\bar{\psi}^{N_\tau}\\ \theta^2&=\psi^2+\bar{\psi}^1 & \tilde{\theta}^2&=\psi^2-\bar{\psi}^1\\ \theta^3&=\psi^3+\bar{\psi}^2 & \tilde{\theta}^3&=\psi^3-\bar{\psi}^2\\ &\vdots & &\vdots\\ \theta^{N_\tau}&=\psi^{N_\tau}+\bar{\psi}^{N_\tau-1} & \tilde{\theta}^{N_\tau}&=\psi^{N_\tau}-\bar{\psi}^{N_\tau-1},\end{aligned} \tag{10}$$

which is encoded in the $2N2N_\tau\times 2N2N_\tau$ matrix U:

$$
\begin{pmatrix} \theta^1 \\ \theta^2 \\ \vdots \\ \theta^{N_\tau} \\ \tilde{\theta}^1 \\ \tilde{\theta}^2 \\ \vdots \\ \tilde{\theta}^{N_\tau} \end{pmatrix} = \underbrace{\left(\begin{array}{cccc|cccc} \mathbb{I}_{2N} & 0 & \cdots & 0 & 0 & \cdots & 0 & -\mathbb{I}_{2N} \\ 0 & \mathbb{I}_{2N} & \cdots & 0 & \mathbb{I}_{2N} & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 & \mathbb{I}_{2N} & \ddots & 0 \\ \vdots & \vdots & & \mathbb{I}_{2N} & 0 & \cdots & \mathbb{I}_{2N} & 0 \\ \hline \mathbb{I}_{2N} & 0 & \cdots & 0 & 0 & \cdots & 0 & \mathbb{I}_{2N} \\ 0 & \mathbb{I}_{2N} & \cdots & 0 & -\mathbb{I}_{2N} & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 & -\mathbb{I}_{2N} & \ddots & 0 \\ \vdots & \vdots & & \mathbb{I}_{2N} & 0 & \cdots & -\mathbb{I}_{2N} & 0 \end{array}\right)}_{:=U} \begin{pmatrix} \psi^1 \\ \psi^2 \\ \vdots \\ \psi^{N_\tau} \\ \bar{\psi}^1 \\ \bar{\psi}^2 \\ \vdots \\ \bar{\psi}^{N_\tau} \end{pmatrix}. \tag{11}
$$

By performing this change of variables we find

$$
\begin{aligned}
W[\phi] = {} & \frac{1}{2^N} \det(U) \int \mathcal{D}\left(\theta, \tilde{\theta}\right)\, \omega(A_1, \theta^1)\, \omega(A_2, \theta^2)\, \omega(A_3, \theta^3) \cdots \omega(A_{N_\tau}, \theta^{N_\tau}) \\
& \times \exp\left(-\frac{1}{2} \begin{bmatrix} \boldsymbol{\theta}^{\mathrm{T}} & \tilde{\boldsymbol{\theta}}^{\mathrm{T}} \end{bmatrix} \left(U^{\mathrm{T}}\right)^{-1} \begin{bmatrix} 0 & -\boldsymbol{M}^{\mathrm{T}} \\ \boldsymbol{M} & 0 \end{bmatrix} U^{-1} \begin{bmatrix} \boldsymbol{\theta} \\ \tilde{\boldsymbol{\theta}} \end{bmatrix}\right) && (12) \\
= {} & \frac{1}{2^N} \det(U) \int \mathcal{D}\left(\theta, \tilde{\theta}\right)\, \omega(A_1, \theta^1)\, \omega(A_2, \theta^2)\, \omega(A_3, \theta^3) \cdots \omega(A_{N_\tau}, \theta^{N_\tau}) \\
& \times \exp\left(-\frac{1}{2} \begin{bmatrix} \boldsymbol{\theta}^{\mathrm{T}} & \tilde{\boldsymbol{\theta}}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{Q} & -\boldsymbol{C}^{\mathrm{T}} \\ \boldsymbol{C} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} \boldsymbol{\theta} \\ \tilde{\boldsymbol{\theta}} \end{bmatrix}\right). && (13)
\end{aligned}
$$

The explicit forms of $\boldsymbol{Q}, \boldsymbol{C}$ and $\boldsymbol{B}$ can be found in Appenix XXXXX. Next we expand the matrix product and split the path integral.

$$
\begin{aligned}
W[\phi] = {} & \frac{1}{2^N} \det(U) \int \mathcal{D}\left(\theta, \tilde{\theta}\right)\, \omega(A_1, \theta^1)\, \omega(A_2, \theta^2)\, \omega(A_3, \theta^3) \cdots \omega(A_{N_\tau}, \theta^{N_\tau}) \\
& \times \exp\left[-\frac{1}{2}\left(\boldsymbol{\theta}^{\mathrm{T}} \boldsymbol{Q} \boldsymbol{\theta} - 2\boldsymbol{\theta}^{\mathrm{T}} \boldsymbol{C}^{\mathrm{T}} \tilde{\boldsymbol{\theta}} + \tilde{\boldsymbol{\theta}}^{\mathrm{T}} \boldsymbol{B} \tilde{\boldsymbol{\theta}}\right)\right] \\
= {} & 2^{(2N_\tau - 1)N} (-1)^{N N_\tau} \int \mathcal{D}(\theta)\, \omega(A_1, \theta^1)\, \omega(A_2, \theta^2)\, \omega(A_3, \theta^3) \cdots \omega(A_{N_\tau}, \theta^{N_\tau}) \exp\left(-\frac{1}{2} \boldsymbol{\theta}^{\mathrm{T}} \boldsymbol{Q} \boldsymbol{\theta}\right) \\
& \times \int \mathcal{D}\left(\tilde{\theta}\right) \exp\left(\frac{1}{2} \tilde{\boldsymbol{\theta}}^{\mathrm{T}} (-\boldsymbol{B})\, \tilde{\boldsymbol{\theta}} + \left(\boldsymbol{\theta}^{\mathrm{T}} \boldsymbol{C}^{\mathrm{T}}\right) \tilde{\boldsymbol{\theta}}\right) \\
= {} & 2^{(2N_\tau - 1)N} (-1)^{N N_\tau} \operatorname{Pf}(-B) \int \mathcal{D}(\theta)\, \omega(A_1, \theta^1)\, \omega(A_2, \theta^2)\, \omega(A_3, \theta^3) \cdots \omega(A_{N_\tau}, \theta^{N_\tau}) \\
& \times \exp\left(-\frac{1}{2} \boldsymbol{\theta}^{\mathrm{T}} \left[\boldsymbol{Q} + \boldsymbol{C}^{\mathrm{T}} \boldsymbol{B}^{-1} \boldsymbol{C}\right] \boldsymbol{\theta}\right), && (14)
\end{aligned}
$$

where $\det(U) = 2^{2NN_\tau}$ was used. In the second equal sign we split the path integral, where we needed to commute the Grassmann differentials which yielded an alternating sign. In the following step we performed the Gaussian integral by using:

$$\int \mathcal{D}(\boldsymbol{\xi}) \exp\left(\frac{1}{2}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{\xi} + \boldsymbol{\zeta}^{T}\boldsymbol{\xi}\right) = \mathrm{Pf}\,(A) \exp\left(\frac{1}{2}\boldsymbol{\zeta}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{\zeta}\right). \tag{15}$$

In the following we assume N_τ to be even. For this case we use $\mathrm{Pf}\,(-B) = (-1)^{\frac{NN_\tau}{2}} \frac{1}{2^{2N(N_\tau-1)}}$ as well as

$$\boldsymbol{R} := -\left[\boldsymbol{Q} + \boldsymbol{C}^{\mathrm{T}}\boldsymbol{B}^{-1}\boldsymbol{C}\right] = \begin{bmatrix} 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \cdots \\ -\mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \cdots \\ \mathbb{I}_{2N} & -\mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & \cdots \\ -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \tag{16}$$

Next we need to find explicit expressions for $\omega(A,\theta)$. In appendix D we show:

$$\omega(A,\theta) = \eta(A) \exp\left(-\frac{1}{2}\boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{G}\boldsymbol{\theta}\right), \tag{17}$$

with

$$\boldsymbol{G} = \frac{\mathbb{I} - \boldsymbol{K}}{\mathbb{I} + \boldsymbol{K}} = \tanh\left(\frac{\boldsymbol{A}}{2}\right), \tag{18}$$

$$\boldsymbol{K} = e^{-\boldsymbol{A}}, \tag{19}$$

$$\eta(\boldsymbol{A}) = (-1)^{N}\,\mathrm{Pf}\left(\begin{bmatrix} \sqrt{2}\,\sinh\left(\frac{\boldsymbol{A}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\,\sinh\left(\frac{\boldsymbol{A}}{4}\right) \end{bmatrix}\right). \tag{20}$$

After plugging this in we find

$$\begin{aligned} W[\phi] &= 2^{N}(-1)^{\frac{NN_\tau}{2}} \left(\prod_{t=1}^{N_\tau} \eta(\boldsymbol{A}_t)\right) \int \mathcal{D}(\boldsymbol{\theta})\,\exp\left[\frac{1}{2}\boldsymbol{\theta}^{\mathrm{T}}\left(\boldsymbol{R} - \mathrm{diag}(\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3, \cdots, \boldsymbol{G}_{N_\tau})\right)\boldsymbol{\theta}\right] \\ &= 2^{N}(-1)^{\frac{NN_\tau}{2}} \left(\prod_{t=1}^{N_\tau} \eta(A_t)\right) \int \mathcal{D}(\boldsymbol{\theta})\,\exp\left[\frac{1}{2}\boldsymbol{\theta}^{\mathrm{T}}\left(\begin{bmatrix} -\boldsymbol{G}_1 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \cdots & -\mathbb{I}_{2N} & \mathbb{I}_{2N} \\ -\mathbb{I}_{2N} & -\boldsymbol{G}_2 & \mathbb{I}_{2N} & \cdots & \mathbb{I}_{2N} & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & -\mathbb{I}_{2N} & -\boldsymbol{G}_3 & \cdots & -\mathbb{I}_{2N} & \mathbb{I}_{2N} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \cdots & -\boldsymbol{G}_{N_\tau-1} & \mathbb{I}_{2N} \\ -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \cdots & -\mathbb{I}_{2N} & -\boldsymbol{G}_{N_\tau} \end{bmatrix}\right)\boldsymbol{\theta}\right], \end{aligned} \tag{21}$$

where we notice that the integral again is of Gaussian form. The final step is to carry out this integration to find

$$W[\phi] = 2^N (-1)^{\frac{NN_\tau}{2}} \left(\prod_{t=1}^{N_\tau} \mathrm{Pf}\left(\begin{bmatrix} \sqrt{2}\sinh\left(\frac{\boldsymbol{A_t}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A_t}}{4}\right) \end{bmatrix} \right) \right)$$
$$\times \mathrm{Pf}\left(\begin{bmatrix} \tanh\left(\frac{\boldsymbol{A}_1}{2}\right) & -\mathbb{I}_{2N} & \cdots & \mathbb{I}_{2N} & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & \tanh\left(\frac{\boldsymbol{A}_2}{2}\right) & \cdots & -\mathbb{I}_{2N} & \mathbb{I}_{2N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \cdots & \tanh\left(\frac{\boldsymbol{A}_{N_\tau-1}}{2}\right) & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \cdots & \mathbb{I}_{2N} & \tanh\left(\frac{\boldsymbol{A}_{N_\tau}}{2}\right) \end{bmatrix} \right). \tag{22}$$

Here we used that for even N_τ the pfaffian does not change if we flip the sign. Now that we have found an analytical closed form of $W[\phi]$ we find for the partition function:

$$Z = \int \mathcal{D}(\phi)\, e^{-S_B[\phi]} W[\phi] = \int \mathcal{D} e^{-S_B[\phi]} (\phi)\, 2^N (-1)^{\frac{NN_\tau}{2}} \left(\prod_{t=1}^{N_\tau} \mathrm{Pf}\left(\begin{bmatrix} \sqrt{2}\sinh\left(\frac{\boldsymbol{A_t}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A_t}}{4}\right) \end{bmatrix} \right) \right)$$
$$\times \mathrm{Pf}\left(\begin{bmatrix} \tanh\left(\frac{\boldsymbol{A}_1}{2}\right) & -\mathbb{I}_{2N} & \cdots & \mathbb{I}_{2N} & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & \tanh\left(\frac{\boldsymbol{A}_2}{2}\right) & \cdots & -\mathbb{I}_{2N} & \mathbb{I}_{2N} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \cdots & \tanh\left(\frac{\boldsymbol{A}_{N_\tau-1}}{2}\right) & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \cdots & \mathbb{I}_{2N} & \tanh\left(\frac{\boldsymbol{A}_{N_\tau}}{2}\right) \end{bmatrix} \right) \tag{23}$$

In this derivation we treated the non-interacting part of the Hamiltonian the same as as the interacting part. This is formally correct, however it is also quite inefficient. In appendix B we analytically integrate out the non-interacting part of the system.

1.1 Reducing time complexity with the caterpillar

The above expression is of time complexity $\mathcal{O}\left(N^3 N_\tau^3\right)$, which is quite inefficient. There is a different formulation of the weight, with a better time complexity. This is in complete analogy to the *sausage* for the determinant. For the pfaffian weight we will refer to it as the *caterpillar*. To see that these two concepts are in complete analogy to one another we actually start out with a quick proof of the *sausage* and then see that the derivation maps over to the pfaffian case in complete analogy. In the determinant case with Dirac fermions we know:

$$\langle\theta|\, e^{\boldsymbol{c}^\dagger \boldsymbol{A} \boldsymbol{c}} |\psi\rangle = e^{\bar{\boldsymbol{\theta}}^{\mathrm{T}} e^{\boldsymbol{A}} \boldsymbol{\psi}} \tag{24}$$

For a product we observe:

$$
\begin{aligned}
\langle\theta|\, e^{\boldsymbol{c}^\dagger \boldsymbol{A}_1 \boldsymbol{c}} e^{\boldsymbol{c}^\dagger \boldsymbol{A}_2 \boldsymbol{c}} \,|\psi\rangle &= \int \mathrm{d}\xi \mathrm{d}\bar{\xi}\, \langle\theta|\exp\left(\boldsymbol{c}^\dagger \boldsymbol{A}_1 \boldsymbol{c}\right)|\xi\rangle \langle\xi|\exp\left(\boldsymbol{c}^\dagger \boldsymbol{A}_2 \boldsymbol{c}\right)|\psi\rangle\, e^{-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}} \\
&= \int \mathrm{d}\xi \mathrm{d}\bar{\xi} \exp\left(\bar{\boldsymbol{\theta}}^{\mathrm{T}} e^{\boldsymbol{A}_1}\boldsymbol{\xi}\right) \exp\left(\bar{\boldsymbol{\xi}}^{\mathrm{T}} e^{\boldsymbol{A}_1}\boldsymbol{\psi}\right) e^{-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}} \\
&= \int \mathrm{d}\xi \mathrm{d}\bar{\xi} \exp\left(-\bar{\boldsymbol{\xi}}\mathbb{I}\boldsymbol{\xi} + \left(\bar{\boldsymbol{\theta}} e^{\boldsymbol{A}_1}\right)\boldsymbol{\xi} + \bar{\boldsymbol{\xi}}^{\mathrm{T}}\left(e^{\boldsymbol{A}_2}\right)\boldsymbol{\psi}\right) \\
&= \exp\left(\bar{\boldsymbol{\theta}}^{\mathrm{T}} e^{\boldsymbol{A}_1} e^{\boldsymbol{A}_2}\boldsymbol{\psi}\right), \qquad (25)
\end{aligned}
$$

where we used Gaussian integration in the last equality. This obviously generalizes to longer products. Now we will evaluate the trace:

$$
\begin{aligned}
\mathrm{Tr}\left\{e^{\boldsymbol{c}^\dagger \boldsymbol{A}_1 \boldsymbol{c}} e^{\boldsymbol{c}^\dagger \boldsymbol{A}_2 \boldsymbol{c}} \ldots e^{\boldsymbol{c}^\dagger \boldsymbol{A}_{N_\tau} \boldsymbol{c}}\right\} &= \int \mathrm{d}\xi \mathrm{d}\bar{\xi}\ \langle-\xi|\, e^{\boldsymbol{c}^\dagger \boldsymbol{A}_1 \boldsymbol{c}} e^{\boldsymbol{c}^\dagger \boldsymbol{A}_2 \boldsymbol{c}} \ldots e^{\boldsymbol{c}^\dagger \boldsymbol{A}_{N_\tau} \boldsymbol{c}}\,|\xi\rangle\, e^{-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}} \\
&= \int \mathrm{d}\xi \mathrm{d}\bar{\xi} \exp\left(-\bar{\boldsymbol{\xi}}^{\mathrm{T}} e^{\boldsymbol{A_1}} e^{\boldsymbol{A_2}} \ldots e^{\boldsymbol{A_{N_\tau}}}\boldsymbol{\xi}\right) e^{-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}} \\
&= \int \mathrm{d}\xi \mathrm{d}\bar{\xi} \exp\left(-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\left(\mathbb{I} + e^{\boldsymbol{A_1}} e^{\boldsymbol{A_2}} \ldots e^{\boldsymbol{A_{N_\tau}}}\right)\boldsymbol{\xi}\right) \\
&= \det\left(\mathbb{I} + e^{\boldsymbol{A_1}} e^{\boldsymbol{A_2}} \ldots e^{\boldsymbol{A_{N_\tau}}}\right), \qquad (26)
\end{aligned}
$$

which is exactly the *sausage*. This procedure can be repeated in complete equivalence for the pfaffian case, with the only difference that the product turns out a more complicated. In appendix D we find

$$
\langle\theta|\, e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\,|\psi\rangle = \eta \exp\left(-\frac{1}{2}\left(\bar{\boldsymbol{\theta}}+\boldsymbol{\psi}\right)^{\mathrm{T}}\boldsymbol{G}\left(\bar{\boldsymbol{\theta}}+\boldsymbol{\psi}\right)\right), \qquad (27)
$$

with

$$
\eta = (-1)^N \,\mathrm{Pf}\left(\begin{bmatrix}\sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right)\end{bmatrix}\right), \qquad \boldsymbol{K} = e^{-\boldsymbol{A}}, \qquad \boldsymbol{G} = \frac{\mathbb{I}-\boldsymbol{K}}{\mathbb{I}+\boldsymbol{K}} \qquad (28)
$$

In the appendix we also derive

$$
\begin{aligned}
&\langle\theta|\, e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)} e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\,|\psi\rangle \qquad (29) \\
&\quad = (-1)^N \eta_1\eta_2 \mathrm{Pf}\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I} \\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix} \exp\left(-\frac{1}{2}\left(\bar{\boldsymbol{\theta}}^{\mathrm{T}}+\boldsymbol{\psi}^{\mathrm{T}}\right)\frac{\mathbb{I}-\boldsymbol{K}_1\boldsymbol{K}_2}{\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2}\left(\bar{\boldsymbol{\theta}}^{\mathrm{T}}+\boldsymbol{\psi}\right)\right) e^{\bar{\boldsymbol{\theta}}^{\mathrm{T}}\boldsymbol{\psi}} \qquad (30)
\end{aligned}
$$

The product of two operator exponentials yields another operator exponential with

$$
\eta_{12} = (-1)^N \eta_1\eta_2 \mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I} \\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right) \qquad \boldsymbol{K}_{12} = \boldsymbol{K}_1\boldsymbol{K}_2 \qquad \boldsymbol{G}_{12} = \frac{\mathbb{I}-\boldsymbol{K}_{12}}{\mathbb{I}+\boldsymbol{K}_{12}} \qquad (31)
$$

For a longer product this generalizes to

$$\langle\theta|e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\dots e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_{N_\tau}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}|\psi\rangle$$
$$= (-1)^{(N_\tau-1)N}\left(\prod_{t=1}^{N_\tau}\eta_t\right)\left(\prod_{t=1}^{N_\tau-1}\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_{1\dots t} & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_{t+1}\end{bmatrix}\right)\right)\exp\left(-\frac{1}{2}\left(\bar{\boldsymbol{\theta}}^{\mathrm{T}}+\bar{\boldsymbol{\psi}}^{\mathrm{T}}\right)\frac{\mathbb{I}-\boldsymbol{K}_1\boldsymbol{K}_2\dots\boldsymbol{K}_{N_\tau}}{\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2\dots\boldsymbol{K}_{N_\tau}}\left(\boldsymbol{\theta}+\boldsymbol{\psi}\right)\right)e^{\bar{\boldsymbol{\theta}}^{\mathrm{T}}\boldsymbol{\psi}}. \tag{32}$$

Now we evaluate the trace:

$$\mathrm{Tr}\left\{e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\dots e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_{N_\tau}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\right\} \tag{33}$$
$$= \int \mathrm{d}\xi\mathrm{d}\bar{\xi}\ \langle-\xi|\,e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\dots e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_{N_\tau}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}|\xi\rangle\,e^{-\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}}$$
$$= \int \mathrm{d}\xi\mathrm{d}\bar{\xi}\ (-1)^{(N_\tau-1)N}\left(\prod_{t=1}^{N_\tau}\eta_t\right)\left(\prod_{t=1}^{N_\tau-1}\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_{1\dots t} & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_{t+1}\end{bmatrix}\right)\right)$$
$$\times\exp\left(-\frac{1}{2}\left(-\bar{\boldsymbol{\xi}}^{\mathrm{T}}+\bar{\boldsymbol{\xi}}^{\mathrm{T}}\right)\frac{\mathbb{I}-\boldsymbol{K}_1\boldsymbol{K}_2\dots\boldsymbol{K}_{N_\tau}}{\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2\dots\boldsymbol{K}_{N_\tau}}\left(-\boldsymbol{\xi}+\boldsymbol{\xi}\right)\right)e^{-2\bar{\boldsymbol{\xi}}^{\mathrm{T}}\boldsymbol{\xi}}. \tag{34}$$

The first exponential obviously vanishes and the rest can be evaluated trivially. We find

$$\mathrm{Tr}\left\{e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\dots e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_{N_\tau}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\right\} \tag{35}$$
$$= (-1)^{(N_\tau+1)N}2^{2N}\left(\prod_{t=1}^{N_\tau}\eta_t\right)\left(\prod_{t=1}^{N_\tau-1}\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_{1\dots t} & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_{t+1}\end{bmatrix}\right)\right). \tag{36}$$

Plugging this into the weight we find the *caterpillar*:

$$W\left[\phi\right] = \frac{1}{2^N}\mathrm{Tr}\left\{e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\dots e^{-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{A}_{N_\tau}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)}\right\}$$
$$= (-1)^{(N_\tau+1)N}2^{N}\left(\prod_{t=1}^{N_\tau}\eta_t\right)\left(\prod_{t=1}^{N_\tau-1}\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_{1\dots t} & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_{t+1}\end{bmatrix}\right)\right)$$
$$= (-1)^{(N_\tau+1)N}\,2^{N}\left(\prod_{t=1}^{N_\tau}(-1)^N\mathrm{Pf}\left(\begin{bmatrix}\sqrt{2}\sinh\left(\frac{\boldsymbol{A}_t}{4}\right) & -\mathbb{I}\\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}_t}{4}\right)\end{bmatrix}\right)\right)\left(\prod_{t=1}^{N_\tau-1}\mathrm{Pf}\left(\begin{bmatrix}\frac{\mathbb{I}-e^{-\boldsymbol{A}_1}e^{-\boldsymbol{A}_2}\dots e^{-\boldsymbol{A}_t}}{\mathbb{I}+e^{-\boldsymbol{A}_1}e^{-\boldsymbol{A}_2}\dots e^{-\boldsymbol{A}_t}} & -\mathbb{I}\\ \mathbb{I} & \frac{\mathbb{I}-e^{-\boldsymbol{A}_{t+1}}}{\mathbb{I}+e^{-\boldsymbol{A}_{t+1}}}\end{bmatrix}\right)\right) \tag{37}$$

This expression is only of time complexity $\mathcal{O}\left(N_\tau N^3\right)$ and is therefore an improvement to the previous expression. However, it should be noted, that this expression is unstable if calculated numerically, while the other formula does not show this instability.

2 Correlators for PfQMC

To run a useful pfQMC alorithm we need a way to calculate observables for a given auxiliary field configuration. Due to Wicks theorem most observables can be calculated from the correlators $\langle \gamma_i(\tau)\gamma_j(0)\rangle$. Here τ labels imaginary time, and i, j label the Majorana operator type as well as a lattice site. Observables can then be build from these correlators. If your for eqxample interested in the Dirac fermion density $c_i^\dagger c_i$ by plugging in the Majorana base $c_i = \frac{1}{2}(\eta_i + i\gamma_i)$ you can find

$$\left\langle c_i^\dagger c_i \right\rangle = \left\langle \frac{1}{2}(1 - i\eta_i\gamma_i) \right\rangle = \frac{1}{2} - i\frac{1}{2}\langle \eta_i \gamma_j\rangle . \tag{38}$$

For a higher correlators you can use a version of Wicks theorem

$$\langle \gamma_a\gamma_b\gamma_c\gamma_d\rangle = \langle \gamma_a\gamma_b\rangle \langle \gamma_c\gamma_d\rangle - \langle \gamma_a\gamma_c\rangle \langle \gamma_b\gamma_d\rangle + \langle \gamma_a\gamma_d\rangle \langle \gamma_b\gamma_c\rangle . \tag{39}$$

You simply need to first remove all pairs of equal operators by use of $\gamma_i^2 = 1$, and then form all possible contractions. For each contraction keep track of the sign by counting the commutations. Odd numbered correlators vanish. What remains is to find an analytical expression for the two body correlator, which is what we do in the now. We start out with

$$\begin{aligned} \langle \gamma_i(\tau)\gamma_j(0)\rangle &= Z^{-1}\mathrm{Tr}\left\{\gamma_i(\tau)\gamma_j(0)e^{-\beta \boldsymbol{H}}\right\} \\ &= Z^{-1}\mathrm{Tr}\left\{e^{\tau\boldsymbol{H}}\gamma_i e^{-\tau\boldsymbol{H}}\gamma_j e^{-\beta\boldsymbol{H}}\right\} \\ &= Z^{-1}\mathrm{Tr}\left\{\gamma_i e^{-\tau\boldsymbol{H}}\gamma_j e^{-(\beta-\tau)\boldsymbol{H}}\right\} \\ &\approx Z^{-1}\mathrm{Tr}\left\{\gamma_i \left[\prod_{t=1}^{l-1} e^{-\Delta_\tau \boldsymbol{H_0}} e^{-\Delta_\tau \boldsymbol{H_1}}\right]\gamma_j \left[\prod_{t=l}^{N_\tau} e^{-\Delta_\tau \boldsymbol{H_0}} e^{-\Delta_\tau \boldsymbol{H_1}}\right]\right\}. \end{aligned} \tag{40}$$

Here we substituted the Heisenberg picture operators, and used the cyclic property of traces. In the last step we used a Suzuki Trotter decomposition of second order (not third like in the improved correlator). We also defined $N_\tau = \frac{\beta}{\Delta_\tau}$ and $l = \frac{\beta-\tau}{\Delta_\tau}$. Next we perform the now familiar Hubbard Stratanovic transformation.

$$\langle \gamma_i(\tau)\gamma_j(0)\rangle = Z^{-1}\int \mathrm{D}(\phi)\, e^{-S_B}\,\mathrm{Tr}\left\{\gamma_i \underbrace{e^{-\frac{1}{4}\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{A_1}\boldsymbol{\gamma}}}_{:=\boldsymbol{U}_1} ... e^{-\frac{1}{4}\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{A_{l-1}}\boldsymbol{\gamma}}\gamma_j e^{-\frac{1}{4}\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{A_l}\boldsymbol{\gamma}} ... e^{-\frac{1}{4}\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{A_{N_\tau}}\boldsymbol{\gamma}}\right\} \tag{41}$$

From the Heisenberg equation of motion we know

$$\boldsymbol{U}_p^{-1}\gamma_i\boldsymbol{U}_p = \sum_m \left[e^{-\boldsymbol{A_p}}\right]_{im}\gamma_m := \sum_m \left[\boldsymbol{B}_p\right]_{im}\gamma_m \tag{42}$$

Therefore one finds

$$\gamma_i \boldsymbol{U}^k = \boldsymbol{U}_k \boldsymbol{U}_k^{-1} \gamma_i \boldsymbol{U}_k = \sum_m [\boldsymbol{B}_k]_{im} \boldsymbol{U}_k \gamma_m \tag{43}$$

Applying this identity to move the two Majorana singe particle operator next to one another one finds

$$\begin{aligned} \langle \gamma_i(\tau)\gamma_j(0)\rangle &= Z^{-1} \int \mathrm{D}\left(\phi\right) e^{-S_B} \sum_m [\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}]_{im} \operatorname{Tr}\left\{\boldsymbol{U}_1 ... \boldsymbol{U}_{l-1} \gamma_m \gamma_j \boldsymbol{U}_l ... \boldsymbol{U}_{N_\tau}\right\} \\ &= Z^{-1} \int \mathrm{D}\left(\phi\right) e^{-S_B} \sum_m [\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}]_{im} \operatorname{Tr}\left\{\gamma_m \gamma_j \boldsymbol{U}_l ... \boldsymbol{U}_{N_\tau} \boldsymbol{U}_1 ... \boldsymbol{U}_{l-1}\right\} \end{aligned} \tag{44}$$

In appendix XXX (ADD THIS, it is also in paper but without derivation) we show

$$\operatorname{Tr}\left\{\gamma_m \gamma_j \boldsymbol{U}_{\pi(1)} ... \boldsymbol{U}_{\pi(N)}\right\} = W[\phi] \left[\frac{\left(\mathbb{I} - \boldsymbol{B}_{\pi(1)} ... \boldsymbol{B}_{\pi(N)}\right)}{\left(\mathbb{I} + \boldsymbol{B}_{\pi(1)} ... \boldsymbol{B}_{\pi(N)}\right)^{-1}}\right]_{mj} = W[\phi] \left[2\left(\mathbb{I} + \boldsymbol{B}_{\pi(1)} ... \boldsymbol{B}_{\pi(N)}\right)^{-1} - \mathbb{I}\right]_{mj}, \tag{45}$$

which is the greens function for this particular operator string. By using this we find

$$\begin{aligned} \langle \gamma_i(\tau)\gamma_j(0)\rangle &= Z^{-1} \int \mathrm{D}\left(\phi\right) e^{-S_B} W[\phi] \sum_m [\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}]_{im} \left[2\left(\mathbb{I} + \boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}\right)^{-1} - \mathbb{I}\right]_{mj} \\ &= Z^{-1} \int \mathrm{D}\left(\phi\right) e^{-S_B} W[\phi] \underbrace{\left[\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} 2\left(\mathbb{I} + \boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}\right)^{-1} - \mathbb{I}\right]_{mj}}_{:=\boldsymbol{C}_{mj}} \end{aligned} \tag{46}$$

The object $\boldsymbol{C}_{mj}$ is what actually needs to be measured. It is possible to rewrite this object in a more convenient form:

$$\begin{aligned} \boldsymbol{C} &= \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \left(2\left(\mathbb{I} + \boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}\right)^{-1} - \mathbb{I}\right) \\ &= \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \left(2\left(\boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \left(\mathbb{I} + \boldsymbol{B}_1 ... \boldsymbol{B}_{N_\tau}\right) \boldsymbol{B}_{N_\tau}^{-1} ... \boldsymbol{B}_l^{-1}\right)^{-1} - \mathbb{I}\right) \\ &= \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \left(2\boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \left(\mathbb{I} + \boldsymbol{B}_1 ... \boldsymbol{B}_{N_\tau}\right)^{-1} \boldsymbol{B}_{N_\tau}^{-1} ... \boldsymbol{B}_l^{-1} - \mathbb{I}\right) \\ &= \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \left(2\boldsymbol{B}_l ... \boldsymbol{B}_{N_\tau} \boldsymbol{M}^{-1} \boldsymbol{B}_{N_\tau}^{-1} ... \boldsymbol{B}_l^{-1} - \mathbb{I}\right) \\ &= \boldsymbol{B}_1 ... \boldsymbol{B}_{N_\tau} 2\boldsymbol{M}^{-1} \boldsymbol{B}_{N_\tau}^{-1} ... \boldsymbol{B}_l^{-1} - \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \\ &= 2\boldsymbol{M}^{-1} \boldsymbol{B}_1 ... \boldsymbol{B}_{N_\tau} \boldsymbol{B}_{N_\tau}^{-1} ... \boldsymbol{B}_l^{-1} - \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \\ &= 2\boldsymbol{M}^{-1} \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} - \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \\ &= \left(2\boldsymbol{M}^{-1} - \mathbb{I}\right) \boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \\ &= \boldsymbol{G}\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1} \end{aligned} \tag{47}$$

Therefore the the correlator is given by

$$\langle\gamma_i(\tau)\gamma_j(0)\rangle = Z^{-1}\int \mathrm{D}\left(\phi\right) e^{-S_B} W\left[\phi\right] \left[\boldsymbol{G}\boldsymbol{B}_1 ... \boldsymbol{B}_{l-1}\right]_{ij} \tag{48}$$

3 pfQMC for the spin-1/2 Kitaev chain

In this section we will apply the previous work to the spin-1/2 Kitaev chain and derive all the quantities needed for an HMC algorithm. We assume a 1D chain with open ends, with the Hamiltonian

$$H = -\sum_{\langle i,j\rangle,\sigma}\left(t\, a_{i,\sigma}^{\dagger} a_{j,\sigma} + \Delta\, a_{i,\sigma}^{\dagger} a_{j,\sigma}^{\dagger} + \text{H.c.}\right) - \mu\sum_{x,\sigma} n_{x,\sigma} + U\sum_{x}\left(n_{x,\uparrow}-\frac{1}{2}\right)\left(n_{x,\downarrow}-\frac{1}{2}\right), \tag{49}$$

where t is the hopping amplitude, U is the onsite interaction, Δ is the pairing amplitude and μ is the chemical potential. To bring this into a more convenient form we gauge transform even site operators such that

$$a_i \rightarrow i a_i \quad \text{and} \quad a_i^{\dagger} \rightarrow -i a_i^{\dagger}. \tag{50}$$

Substituting this we find

$$H = -i\sum_{\langle i,j\rangle,\sigma}\left(t\left(a_{i,\sigma}^{\dagger} a_{j,\sigma} - a_{j,\sigma}^{\dagger} a_{i,\sigma}\right) - \Delta\left(a_{i,\sigma}^{\dagger} a_{j,\sigma}^{\dagger} - a_{j,\sigma} a_{i,\sigma}\right)\right) - \mu\sum_{x,\sigma} n_{x,\sigma} + U\sum_{x}\left(n_{x,\uparrow}-\frac{1}{2}\right)\left(n_{x,\downarrow}-\frac{1}{2}\right), \tag{51}$$

Now we split each fermion into two Majroana fermions:

$$a_i = \frac{1}{2}\left(\eta_i + i\gamma_i\right), \qquad a_i^{\dagger} = \frac{1}{2}\left(\eta_i - i\gamma_i\right) \tag{52}$$

The corresponding commutation relations are those of the Clifford algebra:

$$\{\eta_i,\eta_j\} = \{\gamma_i,\gamma_j\} = 2\delta_{i,j}, \qquad \{\eta_i,\gamma_j\} = 0 \tag{53}$$

After dropping constant terms this leads to

$$H = -\frac{i}{2}\sum_{<i,j>,\sigma}\left[\left(t-\Delta\right)\left(\eta_{i,\sigma}\eta_{j,\sigma}\right) + \left(t+\Delta\right)\left(\gamma_{i,\sigma}\gamma_{j,\sigma}\right)\right] - i\frac{\mu}{2}\sum_{i,\sigma}\eta_{i,\sigma}\gamma_{i,\sigma} - \frac{U}{4}\sum_{i}\eta_{i,\uparrow}\gamma_{i,\uparrow}\eta_{i,\downarrow}\gamma_{i,\downarrow} \tag{54}$$

Now we have multiple choices of how to rewrite and therefore decouple the modes. Namely:

$$\eta_\uparrow\gamma_\uparrow\eta_\downarrow\gamma_\downarrow == 1 + \frac{1}{2}\left(\eta_\uparrow\gamma_\uparrow + \eta_\downarrow\gamma_\downarrow\right)^2 \tag{55}$$

$$\eta_\uparrow\gamma_\uparrow\eta_\downarrow\gamma_\downarrow = -1 - \frac{1}{2}\left(\eta_\uparrow\gamma_\uparrow - \eta_\downarrow\gamma_\downarrow\right)^2 \tag{56}$$

$$\eta_\uparrow\gamma_\uparrow\eta_\downarrow\gamma_\downarrow = -1 - \frac{1}{2}\left(\eta_\uparrow\eta_\downarrow + \gamma_\uparrow\gamma_\downarrow\right)^2 \tag{57}$$

$$\eta_\uparrow\gamma_\uparrow\eta_\downarrow\gamma_\downarrow = +1 + \frac{1}{2}\left(\eta_\uparrow\eta_\downarrow - \gamma_\uparrow\gamma_\downarrow\right)^2 \tag{58}$$

$$\tag{59}$$

Technically there are even more splittings but they are all nonsensical. Even in the above equations, the last two lines decouple the fermion modes, however if there is a finite chemical potential the non interacting Hamiltonian will always couple these two sectors. Here, we focus on the first two equations. Our Hamiltonian then takes the form

$$H = \underbrace{-\frac{i}{2}\sum_{<i,j>,\sigma}\left[(t-\Delta)\left(\eta_{i,\sigma}\eta_{j,\sigma}\right) + (t+\Delta)\left(\gamma_{i,\sigma}\gamma_{j,\sigma}\right)\right] - i\frac{\mu}{2}\sum_{i,\sigma}\eta_{i,\sigma}\gamma_{i,\sigma}}_{:=H_0} \underbrace{\mp\frac{U}{8}\sum_i\left(\eta_{i,\uparrow}\gamma_{i,\uparrow} \pm \eta_{i,\downarrow}\gamma_{i,\downarrow}\right)^2}_{:=H_1}. \tag{60}$$

Now we turn to the corresponding partition function and trotterize:

$$Z = \mathrm{Tr}\left\{e^{-\beta H}\right\} \approx \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H_0} e^{-\Delta_\tau H_1}\right\} \tag{61}$$

$$= \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H_0} e^{\mp\Delta_\tau \frac{U}{8}\sum_i\left(\eta_{i,\uparrow}\gamma_{i,\uparrow} \pm \eta_{i,\downarrow}\gamma_{i,\downarrow}\right)^2}\right\} \tag{62}$$

$$= \int \mathcal{D}(\phi)\, e^{-\frac{2\phi^2}{\Delta_\tau U}}\, \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} \exp\left(-\Delta_\tau H_0\right) \exp\left(-\xi\sum_i \phi_i^t\left(\eta_{i\uparrow}\gamma_{i\uparrow} + \xi^2\eta_{i\downarrow}\gamma_{i\downarrow}\right)\right)\right\}, \tag{63}$$

where in the last line we performed a HS transform, which we kept generic by defining $\xi = 1$ for a real HS transformation and $\xi = i$ for an imaginary HS transformation. What remains is to bring this into a form where we can use the derivations shown in the first section. We therefore define the operator vector:

$$\boldsymbol{\theta}^t := \left(\eta_{\uparrow,1}^t, ..., \eta_{\uparrow,N}^t, \gamma_{\uparrow,1}^t, ..., \gamma_{\uparrow,N}^t, \eta_{\downarrow,1}^t, ..., \eta_{\downarrow,N}^t, \gamma_{\downarrow,1}^t, ..., \gamma_{\downarrow,N}^t,\right)^T \tag{64}$$

With this we can write

$$Z = \int \mathcal{D}(\phi)\, e^{-\frac{2\phi^2}{\Delta_\tau U}}\, \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} \exp\left(-\Delta_\tau \left(\boldsymbol{\theta}^t\right)^T \boldsymbol{H_0}\boldsymbol{\theta}^t\right) \exp\left(-\frac{1}{2}\xi\left(\boldsymbol{\theta}^t\right)^T \boldsymbol{H}_{\mathrm{HS}}\left[\phi^t\right]\boldsymbol{\theta}^t\right)\right\}, \tag{65}$$

where we define

$$\boldsymbol{H}_{\mathrm{HS}}\left[\phi^t\right]=\begin{bmatrix}0 & \boldsymbol{h}^t & 0 & 0\\ -\boldsymbol{h}^t & 0 & 0 & 0\\ 0 & 0 & 0 & \xi^2\boldsymbol{h}^t\\ 0 & 0 & -\xi^2\boldsymbol{h}^t & 0\end{bmatrix}, \qquad \boldsymbol{h}^t=\mathrm{diag}\left(\phi_1^t,...,\phi_N^t\right). \tag{66}$$

Note, the slight abuse of notation. We have gone from H_i being bilinear Majorana operators to $\boldsymbol{H_i}$ being regular matrices which are multiplied by Majorana operator vectors. For the non-interacting Hamiltonian we can find

$$\boldsymbol{H}_0=\begin{bmatrix}\boldsymbol{H}_0^{\uparrow\uparrow} & 0\\ 0 & \boldsymbol{H}_0^{\downarrow\downarrow}\end{bmatrix} \tag{67}$$

$$\boldsymbol{H}_0^{\uparrow\uparrow}=\boldsymbol{H}_0^{\downarrow\downarrow}=-\frac{i}{4}\left[\begin{array}{ccccc|ccccc}0 & \Delta^- & & & & \mu & & & & \\ -\Delta^- & 0 & \Delta^- & & & & \mu & & & \\ & -\Delta^- & 0 & \ddots & & & & \ddots & & \\ & & \ddots & \ddots & \Delta^- & & & & \ddots & \\ & & & -\Delta^- & 0 & & & & & \mu\\ \hline -\mu & & & & & 0 & \Delta^+ & & & \\ & -\mu & & & & -\Delta^+ & 0 & \Delta^+ & & \\ & & \ddots & & & & -\Delta^+ & 0 & \ddots & \\ & & & \ddots & & & & \ddots & \ddots & \Delta^+\\ & & & & -\mu & & & & -\Delta^+ & 0\end{array}\right], \tag{68}$$

where we set $\Delta^{\mp}=(t\mp\Delta)$. We notice that both the non-interaction Hamiltonian as well as the interaction part are block diagonal. The coupling between the two different spin parts has been removed. Therefore as shown in the appendix, the partition function can be rewritten as

$$\begin{aligned}Z=&\int\mathcal{D}\left(\phi\right)\exp\left(-\frac{2\phi^2}{\Delta_\tau U}\right)\mathrm{Tr}\left\{\prod_{t=1}^{N_\tau}\exp\left(-\frac{1}{4}\left(\boldsymbol{\theta}_t^{\uparrow}\right)^T\boldsymbol{A}_{2t-1}^{\uparrow\uparrow}\boldsymbol{\theta}_t^{\uparrow}\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{\theta}_t^{\uparrow}\right)^T\boldsymbol{A}_{2t}^{\uparrow\uparrow}\boldsymbol{\theta}_t^{\uparrow}\right)\right\}\\ &\times\mathrm{Tr}\left\{\prod_{t=1}^{N_\tau}\exp\left(-\frac{1}{4}\left(\boldsymbol{\theta}_t^{\downarrow}\right)^T\boldsymbol{A}_{2t-1}^{\downarrow\downarrow}\boldsymbol{\theta}_t^{\downarrow}\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{\theta}_t^{\downarrow}\right)^T\boldsymbol{A}_{2t}^{\downarrow\downarrow}\boldsymbol{\theta}_t^{\downarrow}\right)\right\} &(69)\\ =&\int\mathcal{D}\left(\phi\right)\exp\left(-\frac{2\phi^2}{\Delta_\tau U}\right)W^{\uparrow}\left[\phi\right]W^{\downarrow}\left[\phi\right] &(70)\end{aligned}$$

where we identify

$$\boldsymbol{A}^{\uparrow\uparrow}_{2t-1} = \boldsymbol{A}^{\downarrow\downarrow}_{2t-1} = 4\Delta_\tau \boldsymbol{H}^{\uparrow\uparrow}_0 \tag{71}$$

$$= -i\Delta_\tau \left[\begin{array}{ccccc|ccccc} 0 & \Delta^- & & & & \mu & & & & \\ -\Delta^- & 0 & \Delta^- & & & & \mu & & & \\ & -\Delta^- & 0 & \ddots & & & & \ddots & & \\ & & \ddots & \ddots & \Delta^- & & & & \ddots & \\ & & & -\Delta^- & 0 & & & & & \mu \\ \hline -\mu & & & & & 0 & \Delta^+ & & & \\ & -\mu & & & & -\Delta^+ & 0 & \Delta^+ & & \\ & & \ddots & & & & -\Delta^+ & 0 & \ddots & \\ & & & \ddots & & & & \ddots & \ddots & \Delta^+ \\ & & & & -\mu & & & & -\Delta^+ & 0 \end{array}\right] \tag{72}$$

and

$$\boldsymbol{A}^{\uparrow\uparrow}_{2t} = \xi^2 \boldsymbol{A}^{\downarrow\downarrow}_{2t} = 2\xi \begin{bmatrix} 0 & \operatorname{diag}(\phi^t_1, \dots, \phi^t_N) \\ -\operatorname{diag}(\phi^t_1, \dots, \phi^t_N) & 0 \end{bmatrix} \tag{73}$$

This is exactly of the form we used in the derivation of the pfaffian weights. By using the results of the main derivation we find

$$\begin{aligned} W^\sigma[\phi] &= (-1)^{(N_\tau+1)N} 2^N \left(\prod_{t=1}^{2N_\tau} \eta_t\right) \left(\prod_{t=1}^{2N_\tau-1} \operatorname{Pf}\left(\begin{bmatrix} \boldsymbol{G}^\sigma_{1\dots t} & -\mathbb{I} \\ \mathbb{I} & \boldsymbol{G}^\sigma_{t+1} \end{bmatrix}\right)\right) \\ &= (-1)^{(2N_\tau+1)N}\, 2^N \left(\prod_{t=1}^{2N_\tau} (-1)^N \operatorname{Pf}\left(\begin{bmatrix} \sqrt{2}\sinh\left(\frac{\boldsymbol{A}^{\sigma\sigma}_t}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}^{\sigma\sigma}_t}{4}\right) \end{bmatrix}\right)\right) \qquad (74) \\ &\times \left(\prod_{t=1}^{2N_\tau-1} \operatorname{Pf}\left(\begin{bmatrix} \frac{\mathbb{I}-e^{-\boldsymbol{A}^{\sigma\sigma}_1} e^{-\boldsymbol{A}^{\sigma\sigma}_2} \dots e^{-\boldsymbol{A}^{\sigma\sigma}_t}}{\mathbb{I}+e^{-\boldsymbol{A}^{\sigma\sigma}_1} e^{-\boldsymbol{A}^{\sigma\sigma}_2} \dots e^{-\boldsymbol{A}^{\sigma\sigma}_t}} & -\mathbb{I} \\ \mathbb{I} & \frac{\mathbb{I}-e^{-\boldsymbol{A}^{\sigma\sigma}_{t+1}}}{\mathbb{I}+e^{-\boldsymbol{A}^{\sigma\sigma}_{t+1}}} \end{bmatrix}\right)\right). \qquad (75) \end{aligned}$$

To simplify this we look at the exponentials of $c \cdot \boldsymbol{A}^{\sigma\sigma}_t$, where c is some complex number. We can rewrite $\boldsymbol{A}^{\sigma\sigma}_{2t}$ as

$$\boldsymbol{A}^{\sigma\sigma}_{2t} = 2\xi \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}_{:=J} \otimes \underbrace{\operatorname{diag}\left(\phi^t_1, \dots, \phi^t_N\right)}_{:=D(\phi^t)}, \qquad \text{with} \quad J^2 = -\mathbb{I}_2. \tag{76}$$

Then we find

$$\exp\left(c\cdot \boldsymbol{A}^{\sigma\sigma}_{2t}\right) = \sum_{n=0} \frac{1}{n!}\left(c\boldsymbol{A}^{\sigma\sigma}_{2t}\right)^n = \sum_n \frac{1}{n!}\left(\boldsymbol{J}\otimes\boldsymbol{D}\left(2\xi c\phi^t\right)\right)^n = \sum_n \frac{1}{n!}\boldsymbol{J}^n\otimes\boldsymbol{D}^n\left(2\xi c\phi^t\right) \tag{77}$$

$$= \mathbb{I}\otimes\sum_n \frac{1}{(2n)!}\boldsymbol{D}^{2n}\left(-2c\xi\phi^t\right) + \boldsymbol{J}\otimes\sum_n \frac{1}{(2n+1)!}\boldsymbol{D}^{2n+1}\left(-2c\xi\phi^t\right) \tag{78}$$

$$= \mathbb{I}\otimes D\left(\cos\left(2c\xi\phi^t\right)\right) + \boldsymbol{J}\otimes\boldsymbol{D}\left(\sin\left(2c\xi\phi^t\right)\right). \tag{79}$$

For latter use we look at

$$\sinh\left(\frac{\boldsymbol{A}^{\sigma\sigma}_{2t}}{4}\right) = J\otimes D\left(\sin\left(\frac{1}{2}\xi\phi^{\mathrm{t}}\right)\right), \tag{80}$$

which we can use for

$$\eta_{2t} = (-1)^N \operatorname{Pf}\left(\begin{bmatrix}\sqrt{2}\sinh\left(\frac{\boldsymbol{A}^{\sigma\sigma}_{2t}}{4}\right) & -\mathbb{I}\\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}^{\sigma\sigma}_{2t}}{4}\right)\end{bmatrix}\right) = (-1)^N \operatorname{Pf}\left(\begin{bmatrix}\sqrt{2}\boldsymbol{J}\otimes\boldsymbol{D}\left(\sin\left(\frac{1}{2}\xi\right)\phi^t\right) & -\mathbb{I}\\ \mathbb{I} & \sqrt{2}\boldsymbol{J}\otimes\boldsymbol{D}\left(\sin\left(\frac{1}{2}\xi\right)\phi^t\right)\end{bmatrix}\right) \tag{81}$$

We define $s_i = \sin\left(\frac{1}{2}\xi\phi^t_i\right)$. By permuting basis vectors the expression can be rewritten as

$$\eta_{2t} = (-1)^N \operatorname{Pf}\left(\bigoplus_{i=1}^N \begin{bmatrix} 0 & \sqrt{2}s_i & -1 & 0\\ -\sqrt{2}s_i & 0 & 0 & -1\\ +1 & 0 & 0 & \sqrt{2}s_i\\ 0 & +1 & -\sqrt{2}s_i & 0\end{bmatrix}\right) = (-1)^N \prod_{i=1}^N \operatorname{Pf}\left(\begin{bmatrix} 0 & \sqrt{2}s_i & -1 & 0\\ -\sqrt{2}s_i & 0 & 0 & -1\\ +1 & 0 & 0 & \sqrt{2}s_i\\ 0 & +1 & -\sqrt{2}s_i & 0\end{bmatrix}\right) \tag{82}$$

$$= (-1)^N \prod_{i=1}^N \left(2\sin^2\left(\frac{1}{2}\xi\phi^t_i\right) - 1\right) = \prod_{i=1}^N \cos\left(\xi\phi^t_i\right), \tag{83}$$

which yields an analytic expression for the first factor in the weight. Next we would like to find an analytic expression for $\boldsymbol{G}_{2t}$. By using similar steps we find

$$\boldsymbol{G}_{2t} = \tanh\left(\frac{A_{2t}}{2}\right) = \left[J\otimes D\left(\sin\left(\xi\phi^t\right)\right)\right]\left[\mathbb{I}\otimes D\left(\cos\left(\xi\phi^t\right)\right)\right]^{-1} \tag{84}$$

$$= \left(J\cdot\mathbb{I}^{-1}\right)\otimes D\left(\sin\left(\xi\phi^t\right)\right)\cdot D\left(\frac{1}{\cos\left(\xi\phi^t\right)}\right) \tag{85}$$

$$= J\otimes D\left(\tan\left(\xi\phi^t\right)\right). \tag{86}$$

3.1 Gradient

For HMC we require the real part of the gradient of the action. One possibility is to take the gradient of the caterpillar and work with its real part. However, there is a more efficient version which relies on the fact, that $W^\sigma = \pm\sqrt{\det\left(M^\sigma\right)}$, where $M^\sigma = \mathbb{I} + B_1^\sigma ... B_{2N_\tau}^\sigma$. From now on we drop the spin index, because the calculation is exactly the same for both sectors. We only need the real part of the gradient of the action, which is the same as taking the derivative of the absolute value of the weight:

$$\mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t} S[\phi]\right) = \mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t} S_B[\phi]\right) - \mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t} \log\left(\mathrm{W}[\phi]\right)\right) = \mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t} S_B[\phi]\right) - \frac{\mathrm{d}}{\mathrm{d}\phi_x^t} \log\left(|\mathrm{W}[\phi]|\right) \quad (87)$$

and the absolute value of the weight is $|W[\phi]| = \left|\sqrt{\det\left(\mathbb{I} + B_1 ... B_{2N_\tau}\right)}\right|$. So it suffices to take the derivative of the square root of the determinant as opposed to the more complicated weight involving pfaffians. We then have

$$\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(-\log\left(|W|\right)\right) = \frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(-\log\left(|\sqrt{\det\left(\mathbb{I} + B_1 B_2 ... B_{2N_\tau}\right)}|\right)\right) \quad (88)$$

$$= -\frac{1}{2}\frac{1}{|W[\phi]|^2}\mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\det\left(\mathbb{I} + B_1 ... B_{2N_\tau}\right)\right) \quad (89)$$

$$= -\frac{1}{2}\mathrm{Re}\left(\mathrm{Tr}\left\{M^{-1}\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(B_1 ... B_{2N_\tau}\right)\right\}\right) \quad (90)$$

$$= -\frac{1}{2}\mathrm{Re}\left(\mathrm{Tr}\left\{M^{-1}B_1 ... \frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(B_{2k-1}B_{2k}\right) ... B_{2N_\tau}\right\}\right) \quad (91)$$

$$= -\frac{1}{2}\mathrm{Re}\left(\mathrm{Tr}\left\{\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(B_{2k-1}B_{2k}\right)B_{2k+1} ... B_{2N_\tau}M^{-1}B_1 ... B_{2k-2}\right\}\right) \quad (92)$$

At this point we need the specific forms of the matrices. We have

$$B_{2t-1} = e^{-A_{2t-1}} := \mathcal{K} \quad (93)$$

which is independent of the auxiliary field ϕ and we have

$$B_{2t} = e^{-A_{2t}} = \mathbb{I} \otimes D\left(\cos\left(2\xi\phi_1^t\right), \ldots, \cos\left(2\xi\phi_N^t\right)\right) - J \otimes D\left(\sin\left(2\xi\phi_1^t\right), \ldots, \sin\left(2\xi\phi_N^t\right)\right). \quad (94)$$

We see that this matrix can be split up into 2x2 block matrices. When we work with indices it will turn out convenient to write matrix indices as spatial indices and block matrix indices. For B_{2t} we can then write:

$$\left[B_{2t}\right]_{(l',f')(l'',f'')} = \delta_{l'l''}\left(c_{l'}^t\left(\delta_{f'1}\delta_{f''1} + \delta_{f'2}\delta_{f''2}\right) - s_{l'}^t\left(\delta_{f'1}\delta_{f''2} - \delta_{f'2}\delta_{f''1}\right)\right) \quad (95)$$

where we defined $s_x^t = \sin\left(2\xi\phi_x^t\right)$ and $c_x^t = \cos\left(2\xi\phi_x^t\right)$. Here l-indices refer to a spatial lattice site and f-indices $\in \{1, 2\}$ refer to the block matrices (refers to types of Majoranas). Then we can calculate the derivative

$$\left[\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}B_{2k}\right]_{(l',f')(l'',f'')} = -2\xi\delta_{l'x}\delta_{l''x}\left[s_x^k\left(\delta_{f'1}\delta_{f''1} + \delta_{f'2}\delta_{f''2}\right) + c_x^k\left(\delta_{f'1}\delta_{f''2} - \delta_{f'2}\delta_{f''1}\right)\right]. \quad (96)$$

Next we look at the product (with Einstein sum convention)

$$\mathcal{K}_{(l,f)(l',f')} \cdot \left[\frac{\mathrm{d}}{\mathrm{d}\phi_x^k} B_{2k}\right]_{(l',f')(l'',f'')} = -2\xi \mathcal{K}_{(l,f)(l',f')} \delta_{l'x} \delta_{l''x} \left[s_x^k \left(\delta_{f'1}\delta_{f''1} + \delta_{f'2}\delta_{f''2}\right) + c_x^k \left(\delta_{f'1}\delta_{f''2} - \delta_{f'2}\delta_{f''1}\right)\right] \quad (97)$$

$$= -2\xi \delta_{l''x} \left(\mathcal{K}_{(l,f)(x,1)} s_x^k \delta_{f''1} + \mathcal{K}_{(l,f)(x,2)} s_x^k \delta_{f''2} + \mathcal{K}_{(l,f)(x,1)} c_x^k \delta_{f''2} - \mathcal{K}_{(l,f)(x,2)} c_x^k \delta_{f''1}\right) \quad (98)$$

With this we are able to calculate the whole gradient:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(-\log\left(|W[\phi]|\right)\right) &= -\frac{1}{2}\mathrm{Re}\left(\mathrm{Tr}\left\{\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(\mathcal{K}B_{2k}\right)\underbrace{\mathcal{K}B_{2k+2}...\mathcal{K}B_{2N_\tau}M^{-1}\mathcal{K}B_2...\mathcal{K}B_{2k-2}}_{:=R^k}\right\}\right) \\
&= \mathrm{Re}\left(\xi\delta_{l''x}\left(\mathcal{K}_{(l,f)(x,1)} s_x^k \delta_{f''1} + \mathcal{K}_{(l,f)(x,2)} s_x^k \delta_{f''2} + \mathcal{K}_{(l,f)(x,1)} c_x^k \delta_{f''2} - \mathcal{K}_{(l,f)(x,2)} c_x^k \delta_{f''1}\right) R^k_{(l'',f'')(l,f)}\right) \\
&= \mathrm{Re}\left(\xi\left(\mathcal{K}_{(l,f)(x,1)} R^k_{(x,1)(l,f)} s_x^k + \mathcal{K}_{(l,f)(x,2)} R^k_{(x,2)(l,f)} s_x^k + \mathcal{K}_{(l,f)(x,1)} R^k_{(x,2)(l,f)} c_x^k - \mathcal{K}_{(l,f)(x,2)} R^k_{(x,1)(l,f)} c_x^k\right)\right) \\
&= \mathrm{Re}\left(\xi\left[\left(R\mathcal{K}\right)_{(x,1)(x,1)} s_x^k + \left(R\mathcal{K}\right)_{(x,2)(x,2)} s_x^k + \left(R\mathcal{K}\right)_{(x,2)(x,1)} c_x^k - \left(R\mathcal{K}\right)_{(x,1)(x,2)} c_x^k\right]\right) \\
&= \mathrm{Re}\left(\xi\mathrm{Tr}_f\left\{\begin{bmatrix} D\left(s^k\right) & D\left(c^k\right) \\ -D\left(c^k\right) & D\left(s^k\right)\end{bmatrix}\begin{bmatrix}\left(R\mathcal{K}\right)_{(1,1)} & \left(R\mathcal{K}\right)_{(1,2)} \\ \left(R\mathcal{K}\right)_{(2,1)} & \left(R\mathcal{K}\right)_{(2,2)}\end{bmatrix}\right\}_{x,x}\right) \\
&= \mathrm{Re}\left(\xi\mathrm{Tr}_f\left\{\underbrace{\begin{bmatrix} D\left(s^k\right) & D\left(c^k\right) \\ -D\left(c^k\right) & D\left(s^k\right)\end{bmatrix} B_{2k}^{-1}}_{=J} B_{2k}\begin{bmatrix}\left(R\mathcal{K}\right)_{(1,1)} & \left(R\mathcal{K}\right)_{(1,2)} \\ \left(R\mathcal{K}\right)_{(2,1)} & \left(R\mathcal{K}\right)_{(2,2)}\end{bmatrix}\right\}_{x,x}\right) \\
&= \mathrm{Re}\left(\xi\mathrm{Tr}_f\left\{\begin{bmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0\end{bmatrix} B_{2k}R\mathcal{K}\right\}_{x,x}\right),
\end{aligned} \quad (99)$$

where Tr_f is to be understood as a trace over the block matrices only. After performing the trace there are only spacial indices left. Let us take a closer look at the result. After resubstituting R^k we find

$$\frac{\mathrm{d}}{\mathrm{d}\phi_x^k}\left(-\log\left(|W[\phi]|\right)\right) = \mathrm{Re}\left[\xi \mathrm{Tr}_f\left\{\begin{bmatrix} 0 & +\mathbb{I} \\ -\mathbb{I} & 0\end{bmatrix} B_{2k}\mathcal{K}B_{2k+2}\mathcal{K}B_{2k+4}...\mathcal{K}B_{2N_\tau}M^{-1}\mathcal{K}B_2...\mathcal{K}B_{2k-2}\mathcal{K}\right\}_{x,x}\right] \tag{100}$$

$$= \mathrm{Re}\left[\xi \mathrm{Tr}_f\left\{\begin{bmatrix} 0 & +\mathbb{I} \\ -\mathbb{I} & 0\end{bmatrix} \underbrace{\mathcal{K}B_{2k+2}\mathcal{K}B_{2k+4}...\mathcal{K}B_{2N_\tau}}_{:=\mathcal{B}(k+1)} M^{-1} \underbrace{\mathcal{K}B_2...\mathcal{K}B_{2k-2}\mathcal{K}B_{2k}}_{:=\mathcal{A}(k)}\right\}_{x,x}\right] \tag{101}$$

$$= \mathrm{Re}\left(\xi \mathrm{Tr}_f\left\{\begin{bmatrix} 0 & +\mathbb{I} \\ -\mathbb{I} & 0\end{bmatrix} \mathcal{B}\left(k+1\right)M^{-1}\mathcal{A}\left(k\right)\right\}_{x,x}\right) \tag{102}$$

$$= \mathrm{Re}\left(\xi\left(\left[\mathcal{B}\left(k+1\right)M^{-1}\mathcal{A}\left(k\right)\right]_{2,1} - \left[\mathcal{B}\left(k+1\right)M^{-1}\mathcal{A}\left(k\right)_{1,2}\right]\right)_{x,x}\right) \tag{103}$$

Here we used the fact that J and B_{2k} commute with one another and the cyclicity of the trace (careful, this is not really rigorous because it is not the full trace, however, it can be shown to be true by defining the x-projection operator and promoting the trace to a full trace and then use cyclicity). Finally, the gradient of the action can be calculated as

$$\begin{aligned}\mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t}S[\phi]\right) &= \mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t}S_B[\phi]\right) - \frac{\mathrm{d}}{\mathrm{d}\phi_x^t}\log\left(|\mathrm{W}[\phi]|\right) \\ &= \mathrm{Re}\left(\frac{\mathrm{d}}{\mathrm{d}\phi_x^t}S_B[\phi]\right) + 2\mathrm{Re}\left(\xi\left(\left[\mathcal{B}\left(k+1\right)M^{-1}\mathcal{A}\left(k\right)\right]_{2,1} - \left[\mathcal{B}\left(k+1\right)M^{-1}\mathcal{A}\left(k\right)_{1,2}\right]\right)_{x,x}\right)\end{aligned} \tag{104}$$

Note that this is not quite general, because we assumed M^σ to be equal to one another, which introduced the factor of two. This result is very similar to the result one finds in DQMC, where one usually assumes a diagonal interaction. It should be noted here however, that the definitions of $\mathcal{A}(k)$ and $\mathcal{B}(k)$ are slightly different from Finns notes.

4 Applying the Pfaffian Formualtion to the Hubbard Model with a super conducting term

In this section we apply the above presented method to a Hubbard Hamiltonian with a superconduction nearest neighbor term. The Hamiltonian takes the from:

$$H = -\sum_{\langle i,j\rangle,\sigma}\left(ta_{i\sigma}^\dagger a_{j\sigma} + \Delta a_{i\sigma}^\dagger a_{j\sigma}^\dagger + \mathrm{H.c.}\right) + U\sum_x\left(n_{x\uparrow} - \frac{1}{2}\right)\left(n_{x\downarrow} - \frac{1}{2}\right) \tag{105}$$

Then we split each Dirac fermion into two Majorana fermions by using:

$$a_{i\sigma} = \frac{1}{2}\left(\eta_{i\sigma} + i\gamma_{i\sigma}\right) \tag{106}$$

$$a_{i\sigma}^{\dagger} = \frac{1}{2}\left(\eta_{i\sigma} - i\gamma_{i\sigma}\right) \tag{107}$$

(Note that this definition is slightly different from the paper from which I took the Hamiltonian. The prefactors are different so the new operators obey the Clifford algebra.) We assume a biparte lattice and gauge transform one sublattice by $a_{i\sigma} \to i a_{i\sigma}$. Then we substitute the Majorana basis and find:

$$H = -\frac{i}{2}\sum_{\langle i,j\rangle\sigma}\left(\left(t+\Delta\right)\gamma_{i\sigma}\gamma_{j\sigma} + \left(t-\Delta\right)\eta_{i\sigma}\eta_{j\sigma}\right) - \frac{U}{4}\sum_{x}\eta_{x\uparrow}\eta_{x\downarrow}\gamma_{x\uparrow}\gamma_{x\downarrow} \tag{108}$$

We need to rewrite the quartic term in a way more suitable for our purposes. To do so we look at

$$\left(\eta_{x\uparrow}\eta_{x\downarrow} \pm \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 = \left(\eta_{x\uparrow}\eta_{x\downarrow}\right)^2 + \left(\gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 \pm 2\eta_{x\uparrow}\eta_{x\downarrow}\gamma_{x\uparrow}\gamma_{x\downarrow} = 2 \pm 2\eta_{x\uparrow}\eta_{x\downarrow}\gamma_{x\uparrow}\gamma_{x\downarrow} \tag{109}$$

Which leads to

$$\eta_{x\uparrow}\eta_{x\downarrow}\gamma_{x\uparrow}\gamma_{x\downarrow} = \frac{1}{2}\left(\eta_{x\uparrow}\eta_{x\downarrow} + \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 - 1 = -\frac{1}{2}\left(\eta_{x\uparrow}\eta_{x\downarrow} - \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 + 1 \tag{110}$$

Substituting this and dropping the constant leads to

$$\begin{aligned} H &= -\frac{i}{2}\sum_{\langle i,j\rangle\sigma}\left(\left(t+\Delta\right)\gamma_{i\sigma}\gamma_{j\sigma} + \left(t-\Delta\right)\eta_{i\sigma}\eta_{j\sigma}\right) - \frac{U}{4}\sum_{x}\left(\eta_{x\uparrow}\eta_{x\downarrow} + \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 && (111) \\ &= -\frac{i}{2}\sum_{\langle i,j\rangle\sigma}\left(\left(t+\Delta\right)\gamma_{i\sigma}\gamma_{j\sigma} + \left(t-\Delta\right)\eta_{i\sigma}\eta_{j\sigma}\right) + \frac{U}{4}\sum_{x}\left(\eta_{x\uparrow}\eta_{x\downarrow} - \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2 := H_K + H_U && (112) \end{aligned}$$

After having brought the Hamiltonian into this convenient form we are interested in the partition function

$$\begin{aligned} Z &= \mathrm{Tr}\left\{e^{-\beta H}\right\} = \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H}\right\} = \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H_K} e^{-\Delta_\tau H_U}\right\} + \mathcal{O}\left(\Delta_\tau^2\right) \\ &= \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H_K}\exp\left(\pm\Delta_\tau\frac{U}{4}\sum_{x}\left(\eta_{x\uparrow}\eta_{x\downarrow} \pm \gamma_{x\uparrow}\gamma_{x\downarrow}\right)^2\right)\right\} + \mathcal{O}\left(\Delta_\tau^2\right) && (113) \end{aligned}$$

Then we perform a Hubbard Stratonovich transformation in each quartic factor:

$$
\begin{aligned}
Z &= \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} e^{-\Delta_\tau H_K}\prod_x \int \mathrm{d}\phi_{tx} e^{\pm\frac{\phi_{tx}^2}{U\Delta_\tau}} \exp\left(-i\phi_{tx}\left(\eta_{x\uparrow}\eta_{x\downarrow} \pm \gamma_{x\uparrow}\gamma_{x\downarrow}\right)\right)\right\} \\
&= \int \mathcal{D}\left(\phi\right)\exp\left(\pm\frac{\phi_{tx}}{U\Delta_\tau}\right)\underbrace{\mathrm{Tr}\left\{\prod_{t=1}^{N_\tau}\exp\left(-\Delta_\tau H_K\right)\exp\left(-i\sum_x \phi_{tx}\left(\eta_{x\uparrow}\eta_{x\downarrow}\pm\gamma_{x\uparrow}\gamma_{x\downarrow}\right)\right)\right\}}_{:=W[\phi]}
\end{aligned} \tag{114}
$$

Now we need to rewrite $W[\phi]$ in terms of bilinear matrix expressions. We define the $4N$-dimensional Majorana operator vector

$$
\boldsymbol{\xi} = \left(\eta_{1\uparrow},\eta_{2\uparrow},\ldots,\eta_{N\uparrow},\eta_{1\downarrow},\eta_{2\downarrow},\ldots,\eta_{N\downarrow},\gamma_{1\uparrow},\gamma_{2\uparrow},\ldots,\gamma_{N,\uparrow},\gamma_{1\downarrow},\gamma_{2\downarrow},\ldots,\gamma_{N\downarrow}\right)^{\mathrm{T}} \tag{115}
$$

Then we can rewrite the weight as

$$
W[\phi] = \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau}\exp\left(-\frac{1}{4}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{\xi}\right)\exp\left(-\frac{1}{4}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{B^{(t)}}\boldsymbol{\xi}\right)\right\}, \tag{116}
$$

where we defined

$$
A = -i\Delta_\tau\begin{pmatrix}(t-\Delta)\boldsymbol{T} & 0 & 0 & 0\\ 0 & (t-\Delta)\boldsymbol{T} & 0 & 0\\ 0 & 0 & (t+\Delta)\boldsymbol{T} & 0\\ 0 & 0 & 0 & (t+\Delta)\boldsymbol{T}\end{pmatrix}, \qquad \boldsymbol{B}^{(t)} = 2i\begin{pmatrix}0 & \boldsymbol{B}_\phi^{(t)} & 0 & 0\\ -\boldsymbol{B}_\phi^{(t)} & 0 & 0 & 0\\ 0 & 0 & 0 & \pm\boldsymbol{B}_\phi^{(t)}\\ 0 & 0 & \mp\boldsymbol{B}_\phi^{(t)} & 0\end{pmatrix} \tag{117}
$$

with

$$
\boldsymbol{T}_{ij} = \begin{cases}+1, & \text{if } i<j \text{ and } i,j \text{ n.n.},\\ -1, & \text{if } i>j \text{ and } i,j \text{ n.n.},\\ 0, & \text{otherwise,}\end{cases} \qquad \boldsymbol{B}_\phi^{(t)} = \mathrm{diag}\left(\phi_1^{(t)},\phi_2^{(t)},\ldots,\phi_N^{(t)}\right). \tag{118}
$$

Now $W[\phi]$ is in the form that we require and we can rewrite the partition function as

$$
\begin{aligned}
Z = &\int \mathcal{D}(\phi) \exp\left(\pm\frac{\phi^2}{\Delta_\tau U}\right)(-1)^{NN_\tau}\, 2^{2N} \\
&\times \prod_{t=1}^{N_\tau} \mathrm{Pf}\left(\begin{bmatrix} \sqrt{2}\mathrm{sinh}\left(\frac{e^{\boldsymbol{A}}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\mathrm{sinh}\left(\frac{\boldsymbol{A}}{4}\right)\end{bmatrix}\right) \mathrm{Pf}\left(\begin{bmatrix} \sqrt{2}\mathrm{sinh}\left(\frac{\boldsymbol{B}^{(t)}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\mathrm{sinh}\left(\frac{\boldsymbol{B}^{(t)}}{4}\right)\end{bmatrix}\right) \\
&\times \mathrm{Pf}\left(\begin{bmatrix} \tanh\left(\frac{\boldsymbol{A}}{2}\right) & -\mathbb{I} & \mathbb{I} & -\mathbb{I} & \dots & -\mathbb{I} \\ \mathbb{I} & \tanh\left(\frac{\boldsymbol{B}^{(1)}}{2}\right) & -\mathbb{I} & \mathbb{I} & & \vdots \\ -\mathbb{I} & \mathbb{I} & \tanh\left(\frac{\boldsymbol{A}}{2}\right) & -\mathbb{I} & & \vdots \\ \mathbb{I} & -\mathbb{I} & \mathbb{I} & \tanh\left(\frac{\boldsymbol{B}^{(2)}}{2}\right) & & \vdots \\ \vdots & & & & \ddots & \vdots \\ \mathbb{I} & \dots & \dots & \dots & \dots & \tanh\left(\frac{\boldsymbol{B}^{(N_\tau)}}{2}\right) \end{bmatrix}\right),
\end{aligned}
\tag{119}
$$

Which branch you choose depends on the sign of U.
We se that the matrices $\boldsymbol{A}$ and $\boldsymbol{B}^{(t)}$ are block diagonal with

$$
\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_\eta & 0 \\ 0 & \boldsymbol{A}_\gamma \end{bmatrix} \qquad \boldsymbol{B}^{(t)} = \begin{bmatrix} \boldsymbol{B}_\eta^{(t)} & 0 \\ 0 & \boldsymbol{B}_\gamma^{(t)} \end{bmatrix}
\tag{120}
$$

In appendix A we derive that in this case the result can be factorized. Applying this we find

$$
\begin{aligned}
Z = &\int \mathcal{D}(\phi) \exp\left(\pm\frac{\phi^2}{4U\Delta_\tau}\right) 2^{2N}(-1)^{NN_\tau} \\
&\times \left(\prod_{t=1}^{N_\tau} \eta\left(\boldsymbol{A}_\eta\right)\eta\left(\boldsymbol{B}_\eta^{(t)}\right)\right) \mathrm{Pf}\left(\begin{bmatrix} \tanh\left(\frac{\boldsymbol{A}_\eta}{2}\right) & -\mathbb{I} & \dots & -\mathbb{I} \\ \mathbb{I} & \tanh\left(\frac{\boldsymbol{B}_\eta^{(1)}}{2}\right) & & \vdots \\ \vdots & & \ddots & \vdots \\ \mathbb{I} & \dots & \dots & \tanh\left(\frac{\boldsymbol{B}_\eta^{(N_\tau)}}{2}\right) \end{bmatrix}\right) \\
&\times \left(\prod_{t=1}^{N_\tau} \eta\left(\boldsymbol{A}_\gamma\right)\eta\left(\pm\boldsymbol{B}_\gamma^{(t)}\right)\right) \mathrm{Pf}\left(\begin{bmatrix} \tanh\left(\frac{\boldsymbol{A}_\gamma}{2}\right) & -\mathbb{I} & \dots & -\mathbb{I} \\ \mathbb{I} & \tanh\left(\frac{\pm\boldsymbol{B}_\gamma^{(1)}}{2}\right) & & \vdots \\ \vdots & & \ddots & \vdots \\ \mathbb{I} & \dots & \dots & \tanh\left(\frac{\pm\boldsymbol{B}_\gamma^{(N_\tau)}}{2}\right) \end{bmatrix}\right),
\end{aligned}
\tag{121}
$$

where the signs are again correlated. One should notice that here unfortunately the notations slightly clash, as the Majorana operators and the coefficient of the Gaussian are both labeled by η. It is noted, that whenever η is written with an argument, we mean the coefficient of the Gaussian.

A Factorization of the expression for Block Diagonal matrices

If we have 2-block×2-block diagonal matrices of the form

$$\boldsymbol{A}^{(t)} = \begin{bmatrix} \boldsymbol{B}^{(t)} & 0 \\ 0 & \bar{\boldsymbol{B}}^{(t)} \end{bmatrix} \tag{122}$$

in the weight we can write

$$W[\phi] = \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} \exp\left(-\frac{1}{4}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{A}^{(t)}\boldsymbol{\xi}\right)\right\} = \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} \exp\left(-\frac{1}{4}\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{B}^{(t)}\boldsymbol{\gamma}\right)\exp\left(-\frac{1}{4}\boldsymbol{\eta}^{\mathrm{T}}\bar{\boldsymbol{B}}^{(t)}\boldsymbol{\eta}\right)\right\}, \tag{123}$$

where we defined the $4N$-dimensional vector $\boldsymbol{\xi} = \begin{bmatrix}\boldsymbol{\gamma} & \boldsymbol{\eta}\end{bmatrix}^{\mathrm{T}}$. Continuing as before we enlarge the Hilbert space to find

$$\begin{aligned}
W[\phi] &= \mathrm{Tr}\left\{\prod_{t=1}^{N_\tau} \exp\left(-\frac{1}{4}(\boldsymbol{f}^\dagger + \boldsymbol{f})^{\mathrm{T}}\boldsymbol{B}^{(t)}(\boldsymbol{f}^\dagger + \boldsymbol{f})\right)\exp\left(-\frac{1}{4}(\boldsymbol{g}^\dagger + \boldsymbol{g})^{\mathrm{T}}\bar{\boldsymbol{B}}^{(t)}(\boldsymbol{g}^\dagger + \boldsymbol{g})\right)\right\} \\
&= \frac{1}{2^{2N}}\int \mathcal{D}\left(\psi,\bar{\psi},\phi,\bar{\phi}\right) \\
&\quad \times \langle -\psi^{N_\tau}, -\phi^{N_\tau}|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{B}^{(1)}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)^{\mathrm{T}}\bar{\boldsymbol{B}}^{(1)}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)\right)|\psi^1,\phi^1\rangle \\
&\quad \times \langle -\psi^{1}, -\phi^{1}|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{B}^{(2)}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)^{\mathrm{T}}\bar{\boldsymbol{B}}^{(2)}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)\right)|\psi^2,\phi^2\rangle \\
&\quad \vdots \\
&\quad \times \langle -\psi^{N_\tau-1}, -\phi^{N_\tau-1}|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}}\boldsymbol{B}^{(N_\tau)}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)^{\mathrm{T}}\bar{\boldsymbol{B}}^{(N_\tau)}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)\right)|\psi^{N_\tau},\phi^{N_\tau}\rangle \\
&\quad \times \exp\left(-\bar{\boldsymbol{\psi}}^{\mathrm{T}}\boldsymbol{\psi}\right)\exp\left(-\bar{\boldsymbol{\phi}}^{\mathrm{T}}\boldsymbol{\phi}\right)
\end{aligned} \tag{124}$$

where $|\psi,\phi\rangle = |\psi\rangle \otimes |\phi\rangle$. The operator $\boldsymbol{f}$ ($\boldsymbol{g}$) only acts on $|\psi\rangle$ ($|\phi\rangle$). Using this we can factorize the above expression to

$$
\begin{aligned}
W[\phi] = \frac{1}{2^N} \int \mathcal{D}\left(\psi, \bar{\psi}\right) &\langle -\psi^{N_\tau} | \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}} \boldsymbol{B^{(1)}} \left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right) |\psi^1\rangle \\
&\langle -\psi^1 | \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}} \boldsymbol{B^{(2)}} \left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right) |\psi^2\rangle \\
&\vdots \\
&\langle \psi^{N_\tau - 1} | \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}} \boldsymbol{B^{(N_\tau)}} \left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right) |\psi^{N_\tau}\rangle \exp\left(-\bar{\boldsymbol{\psi}}^{\mathrm{T}} \boldsymbol{\psi}\right) \\
\times \frac{1}{2^N} \int \mathcal{D}\left(\psi, \bar{\psi}\right) &\langle -\psi^{N_\tau} | \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)^{\mathrm{T}} \boldsymbol{B^{(1)}} \left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right) |\psi^1\rangle \\
&\langle -\phi^1 | \exp\left(-\frac{1}{4}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)^{\mathrm{T}} \bar{\boldsymbol{B}}^{\boldsymbol{(2)}} \left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)\right) |\phi^2\rangle \\
&\vdots \\
&\langle \phi^{N_\tau - 1} | \exp\left(-\frac{1}{4}\left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)^{\mathrm{T}} \bar{\boldsymbol{B}}^{\boldsymbol{(N_\tau)}} \left(\boldsymbol{g}^\dagger + \boldsymbol{g}\right)\right) |\phi^{N_\tau}\rangle \exp\left(-\bar{\boldsymbol{\phi}}^{\mathrm{T}} \boldsymbol{\phi}\right)
\end{aligned}
\tag{125}
$$

From here each path integral can be computed as presented in the main text.

$$
W[\phi] = W_1[\phi] W_2[\phi] \tag{126}
$$

Therefore, if the matrices in the string of exponentials are block diagonal, the entire expression can be factorized into a product over multiple weights. However, the two weights always share the same auxiliary field configuration. This factorization is sometimes useful to reduce the sign problem.

B Integrating out the non-interacting part of the Hamiltonian

Here, we will attempt to analytically integrate out the factors in the operator string which do not depend on some auxiliary field. Let us assume the Hamiltonian after some Hubbard-Stratanovich transformation can be written as

$$
H = H_0 + H_{\mathrm{int}}[\phi], \tag{127}
$$

then after a Suzuki-Trotter approximation the weight can be written as

$$
W[\phi] \approx \mathrm{Tr} \left\{ \prod_{t=0}^{N_\tau} \exp\left(-\frac{1}{4} \boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{A} \boldsymbol{\gamma}\right) \exp\left(-\frac{1}{4} \boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{B}^{(t)} \boldsymbol{\gamma}\right) \right\} \tag{128}
$$

Most of the calculation is very similar to the main text, therefore we will go through some of the calculations much quicker. We proceed again by enlargening the Hilbert space. Then we insert $2N_\tau$

completeness relations for the coherent basis - one between each exponential factor. Then by using the same identities as before we can again integrate out the non-physical degrees of freedom. Eventually we end up with

$$W[\phi] = 2^N (-1)^{NN_\tau} \int \mathcal{D}(\theta)\, \omega(\boldsymbol{A}, \theta^1)\omega(\boldsymbol{A}, \theta^3)\dots\omega(\boldsymbol{A}, \theta^{2N_\tau-1})\omega(\boldsymbol{B}^{(1)}, \theta^2)\omega(\boldsymbol{B}^{(2)}, \theta^4)\dots\omega(\boldsymbol{B}^{(N_\tau)}, \theta^{2N_\tau}) \\ \times \exp\left(\frac{1}{2}\boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{\theta}\right). \tag{129}$$

We notice that all the odd numbered Grassmann variables belong to expressions independent of the auxiliary field. We therefore expand the product in the exponent and split the path integral.

$$\begin{aligned} W[\phi] &= (-1)^{NN_\tau}\, 2^N \int \mathcal{D}\left(\theta^{\mathrm{even}}, \theta^{\mathrm{odd}}\right) \omega(\boldsymbol{A}, \theta^1)\omega(\boldsymbol{A}, \theta^3)\dots\omega(\boldsymbol{A}, \theta^{2N_\tau-1})\omega(\boldsymbol{B}^{(1)}, \theta^2)\omega(\boldsymbol{B}^{(2)}, \theta^4)\dots\omega(\boldsymbol{B}^{(N_\tau)}, \theta^{2N_\tau}) \\ &\qquad \times \exp\left(\frac{1}{2}\begin{bmatrix}\boldsymbol{\theta}^{\mathrm{odd}} & \boldsymbol{\theta}^{\mathrm{even}}\end{bmatrix}\boldsymbol{R'}\begin{bmatrix}\boldsymbol{\theta}^{\mathrm{odd}} \\ \boldsymbol{\theta}^{\mathrm{even}}\end{bmatrix}\right) \\ &= (-1)^{NN_\tau}\, 2^N \int \mathcal{D}\left(\theta^{\mathrm{even}}\right) \omega(\boldsymbol{B}^{(1)}, \theta^2)\omega(\boldsymbol{B}^{(2)}, \theta^4)\dots\omega(\boldsymbol{B}^{(N_\tau)}, \theta^{2N_\tau}) \exp\left(\frac{1}{2}\left(\boldsymbol{\theta}^{\mathrm{even}}\right)^{\mathrm{T}}\boldsymbol{R}_{22}\boldsymbol{\theta}^{\boldsymbol{even}}\right) \\ &\qquad \times \int \mathcal{D}\left(\theta^{\boldsymbol{odd}}\right) \omega(\boldsymbol{A}, \theta^1)\omega(\boldsymbol{A}, \theta^3)\dots\omega(\boldsymbol{A}, \theta^{2N_\tau-1}) \exp\left(\frac{1}{2}\left(\boldsymbol{\theta}^{\mathrm{odd}}\right)^{\mathrm{T}}\boldsymbol{R'}_{11}\boldsymbol{\theta}^{\mathrm{odd}} + \left(\boldsymbol{\theta}^{\mathrm{even}}\right)^{\mathrm{T}}\boldsymbol{R'}_{21}\boldsymbol{\theta}^{\mathrm{odd}}\right), \end{aligned} \tag{130}$$

where we have permuted the basis vectors so that all the odd once appear first and then the even ones follow. This change of basis of course also changes the matrix $\boldsymbol{R}$ to $\boldsymbol{R'}$. Written out this leads to

$$\boldsymbol{R'}_{11} = \begin{bmatrix} 0 & -\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \dots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & & \vdots \\ \mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & \dots & \dots & \mathbb{I}_{2N} & 0 \end{bmatrix} = \boldsymbol{R'}_{22} \tag{131}$$

and

$$\boldsymbol{R'}_{21} = \begin{bmatrix} -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \mathbb{I}_{2N} & \dots & \mathbb{I}_{2N} \\ -\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & & \vdots \\ -\mathbb{I}_{2N} & -\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \mathbb{I}_{2N} \\ -\mathbb{I}_{2N} & \dots & \dots & -\mathbb{I}_{2N} & -\mathbb{I}_{2N} \end{bmatrix} = -\boldsymbol{R'}^{\mathrm{T}}_{21} \tag{132}$$

Now we substitute the expression for $\omega(A, \theta^{2t-1})$, which leads to another Gaussian integral:

$$W[\phi] = (-1)^{NN_\tau}\, 2^N \left(\eta\left(\boldsymbol{A}\right)\right)^{N_\tau} \int \mathcal{D}\left(\theta^{\text{even}}\right) \omega(\boldsymbol{B}^{(1)}, \theta^2)\omega(\boldsymbol{B}^{(4)}, \theta^4)\ldots\omega(\boldsymbol{B}^{(2N_\tau)}, \theta^{2N_\tau}) \exp\left(\frac{1}{2}\left(\boldsymbol{\theta}^{\text{even}}\right)^{\text{T}} \boldsymbol{R}_{22}\boldsymbol{\theta}^{even}\right)$$

$$\times \int \mathcal{D}\left(\boldsymbol{\theta}^{odd}\right) \omega(\boldsymbol{A}, \theta^1)\omega(\boldsymbol{A}, \theta^3)\ldots\omega(\boldsymbol{A}, \theta^{2N_\tau - 1}) \exp\left(\frac{1}{2}\left(\boldsymbol{\theta}^{\text{odd}}\right)^{\text{T}} \underbrace{\left[\boldsymbol{R'}_{11} - \left(\mathbb{I}_{N_\tau} \otimes \boldsymbol{A}\right)\right]}_{:=\boldsymbol{K}} \boldsymbol{\theta}^{\text{odd}} + \left(\boldsymbol{\theta}^{\text{even}}\right)^{\text{T}} \boldsymbol{R'}_{21}\boldsymbol{\theta}^{\text{odd}}\right)$$

$$= (-1)^{NN_\tau}\, 2^N \left(\eta\left(\boldsymbol{A}\right)\right)^{N_\tau} \text{Pf}\left(\boldsymbol{K}\right) \int \mathcal{D}\left(\theta^{\text{even}}\right) \omega(\boldsymbol{B}^{(1)}, \theta^2)\omega(\boldsymbol{B}^{(4)}, \theta^4)\ldots\omega(\boldsymbol{B}^{(2N_\tau)}, \theta^{2N_\tau})$$

$$\times \exp\left(\frac{1}{2}\left(\boldsymbol{\theta}^{\text{even}}\right)^{\text{T}} \left[\boldsymbol{R}_{22} + \boldsymbol{R'}_{21}\boldsymbol{K}^{-1}\boldsymbol{R'}^{\text{T}}_{21}\right] \boldsymbol{\theta}^{\text{even}}\right). \tag{133}$$

Substituting the expressions for $\omega(\boldsymbol{B}^{(t)}, \theta^{(2t)})$ and carrying out the integration leads to

$$W[\phi] = (-1)^{NN_\tau}\, 2^N \left(\eta\left(\boldsymbol{A}\right)\right)^{N_\tau} \text{Pf}\left(\boldsymbol{K}\right) \left(\prod_{t=1}^{N_\tau} \eta\left(\boldsymbol{B}^{(t)}\right)\right) \text{Pf}\left(\boldsymbol{R'}_{22} + \boldsymbol{R'_{21}}\boldsymbol{K^{-1}}\boldsymbol{R'^{T}_{21}} - \text{diag}\left(\boldsymbol{G}_1, \boldsymbol{G}_2, \ldots, \boldsymbol{G}_{N_\tau}\right)\right) \tag{134}$$

This result seems quite nice, however the argument of the pfaffian seems quite complicated. As of right now I was unable to find a nice closed form solution for this matrix. Perhaps this must be handled by the computer later on. It does not seem like the matrix needs to be calculated often.

C Calculation of large matrices

In this section we will show which matrices appear in the derivations. First we list some of the matrices that were defined and write some of them in a more convenient form:

$$\boldsymbol{M} := \begin{bmatrix} \mathbb{I}_{2N} & -\mathbb{I}_{2N} & 0 & \cdots & 0 \\ 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \mathbb{I}_{2N} & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & \cdots & \cdots & 0 & \mathbb{I}_{2N} \end{bmatrix} \tag{135}$$

$$\boldsymbol{U} = \left[\begin{array}{cccc|cccc} \mathbb{I}_{2N} & 0 & \cdots & 0 & 0 & \cdots & 0 & -\mathbb{I}_{2N} \\ 0 & \mathbb{I}_{2N} & \cdots & 0 & \mathbb{I}_{2N} & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 & \mathbb{I}_{2N} & \ddots & 0 \\ \vdots & \vdots & & \mathbb{I}_{2N} & 0 & \cdots & \mathbb{I}_{2N} & 0 \\ \hline \mathbb{I}_{2N} & 0 & \cdots & 0 & 0 & \cdots & 0 & \mathbb{I}_{2N} \\ 0 & \mathbb{I}_{2N} & \cdots & 0 & -\mathbb{I}_{2N} & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 & -\mathbb{I}_{2N} & \ddots & 0 \\ \vdots & \vdots & & \mathbb{I}_{2N} & 0 & \cdots & -\mathbb{I}_{2N} & 0 \end{array}\right] := \begin{bmatrix} \mathbb{I} & \boldsymbol{\rho} \\ \mathbb{I} & -\boldsymbol{\rho} \end{bmatrix}. \tag{136}$$

One immediately sees

$$\boldsymbol{U}^{-1} = \frac{1}{2}\boldsymbol{U}^{\mathrm{T}}, \qquad \text{and} \qquad \left(\boldsymbol{U}^{\mathrm{T}}\right)^{-1} = \frac{1}{2}\boldsymbol{U}. \tag{137}$$

To calculate the determinant of $\boldsymbol{U}$ we use

$$\det(\boldsymbol{U}) = \det(\mathbb{I}) \det\left(-\boldsymbol{\rho} - \mathbb{III}^{-1}\boldsymbol{\rho}\right) = \det(2\boldsymbol{\rho}) = 2^{2NN_\tau}. \tag{138}$$

In the text we used

$$\begin{bmatrix} \boldsymbol{Q} & -\boldsymbol{C}^{\mathrm{T}} \\ \boldsymbol{C} & \boldsymbol{B} \end{bmatrix} = \left(\boldsymbol{U}^{\mathrm{T}}\right)^{-1} \begin{bmatrix} 0 & -\boldsymbol{M}^{\mathrm{T}} \\ \boldsymbol{M} & 0 \end{bmatrix} \boldsymbol{U}^{-1}. \tag{139}$$

Then

$$\boldsymbol{Q} = \frac{1}{4} \begin{bmatrix} 0 & -\mathbb{I}_{2N} & & & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & & \\ & \mathbb{I}_{2N} & 0 & \ddots & \\ & & \ddots & \ddots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & & & \mathbb{I}_{2N} & 0 \end{bmatrix} = -\boldsymbol{B} \tag{140}$$

$$\boldsymbol{C} = \frac{1}{4} \begin{bmatrix} 2\mathbb{I}_{2N} & -\mathbb{I}_{2N} & & & \mathbb{I}_{2N} \\ -\mathbb{I}_{2N} & 2\mathbb{I}_{2N} & -\mathbb{I}_{2N} & & \\ & -\mathbb{I}_{2N} & 2\mathbb{I}_{2N} & \ddots & \\ & & \ddots & \ddots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & & & -\mathbb{I}_{2N} & 2\mathbb{I}_{2N} \end{bmatrix} \tag{141}$$

must hold. It turns out that

$$\boldsymbol{B}^{-1} = 2 \begin{bmatrix} 0 & -\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \dots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & 0 & \ddots & 0 \\ 0 & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \ddots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & 0 & \ddots & 0 \\ 0 & \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \ddots & -\mathbb{I}_{2N} \\ \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & \mathbb{I}_{2N} & 0 & \dots & 0 \end{bmatrix}. \tag{142}$$

Next we find

$$
\boldsymbol{B}^{-1}\boldsymbol{C} = \begin{bmatrix}
0 & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \dots & -\mathbb{I}_{2N} \\
\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \ddots & \mathbb{I}_{2N} \\
-\mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \ddots & -\mathbb{I}_{2N} \\
\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \ddots & \mathbb{I}_{2N} \\
-\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & \ddots & -\mathbb{I}_{2N} \\
\vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\
\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \dots & 0
\end{bmatrix} \tag{143}
$$

and

$$
\boldsymbol{C}^{\mathrm{T}}\boldsymbol{B}^{-1}\boldsymbol{C} = \frac{1}{4}\begin{bmatrix}
0 & -3\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & \dots & 3\mathbb{I}_{2N} \\
3\mathbb{I}_{2N} & 0 & -3\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & \ddots & 4\mathbb{I}_{2N} \\
-4\mathbb{I}_{2N} & 3\mathbb{I}_{2N} & 0 & -3\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & \ddots & -4\mathbb{I}_{2N} \\
4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 3\mathbb{I}_{2N} & 0 & -3\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & \ddots & 4\mathbb{I}_{2N} \\
-4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 3\mathbb{I}_{2N} & 0 & -3\mathbb{I}_{2N} & \ddots & -4\mathbb{I}_{2N} \\
4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 3\mathbb{I}_{2N} & 0 & \ddots & 4\mathbb{I}_{2N} \\
\vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\
3\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & 4\mathbb{I}_{2N} & -4\mathbb{I}_{2N} & \dots & 0
\end{bmatrix}. \tag{144}
$$

And finally we obtain

$$
\boldsymbol{R} = -\left(\boldsymbol{Q} + \boldsymbol{C}^{\mathbf{T}}\boldsymbol{B}^{-1}\boldsymbol{C}\right) = \begin{bmatrix}
0 & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \dots & -\mathbb{I}_{2N} \\
\mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & \ddots & \mathbb{I}_{2N} \\
-\mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & -\mathbb{I}_{2N} & \ddots & -\mathbb{I}_{2N} \\
\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & 0 & \ddots & \mathbb{I}_{2N} \\
\vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\
\mathbb{I}_{2N} & -\mathbb{I}_{2N} & \mathbb{I}_{2N} & -\mathbb{I}_{2N} & \dots & 0
\end{bmatrix}. \tag{145}
$$

D Deriving the Gaussian Form of the Matrix elements

We are aiming to derive

$$
\omega\left(A, \theta\right) = \eta(A) \exp\left(-\frac{1}{2}\boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{G}\boldsymbol{\theta}\right), \tag{146}
$$

with

$$\boldsymbol{G} = \frac{\mathbb{I} - \boldsymbol{K}}{\mathbb{I} + \boldsymbol{K}} = \tanh\left(\frac{\boldsymbol{K}}{2}\right), \tag{147}$$

$$\boldsymbol{K} = e^{-\boldsymbol{A}}, \tag{148}$$

$$\eta(\boldsymbol{A}) = (-1)^N \operatorname{Pf}\left(\begin{bmatrix} \sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right) & -\mathbb{I} \\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right)\end{bmatrix}\right). \tag{149}$$

To do so, we will use the following strategy. First we show that for very small exponents the relation holds. Then we show, that if the relation holds for some matrices A_1 and A_2, it must also hold for the product of their exponentials. We see how $\boldsymbol{G}$ and η behave under this multiplication and can then derive a set of differential equations to find the form of these quantities. This is equivalent to looking at how a string of products of exponentials with small arguments behaves.
We look at

$$\begin{aligned}\langle\theta| \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\Delta_x \boldsymbol{A}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right)|\psi\rangle &= \langle\theta|\left(1 - \frac{\Delta_x}{4}\sum_{ij}\left(f_i^\dagger + f_i\right)A_{ij}\left(f_j^\dagger + f_j\right)\right)|\psi\rangle + \mathcal{O}\left(\Delta_x^2\right) \\ &= \langle\theta|\psi\rangle - \frac{\Delta_x}{4}\langle\theta|\sum_{ij}\left(f_i^\dagger f_j^\dagger + f_i^\dagger f_j + f_i f_j^\dagger + f_i f_j\right)A_{ij}|\psi\rangle + \mathcal{O}\left(\Delta_x^2\right)\end{aligned} \tag{150}$$

Due to the anti commutation relations for Majorana fermions only the anti symmetric part of $\boldsymbol{A}$ gives a contribution (the diagonal parts technically also give a contribution but only as a constant shift in the hamiltonian which is equivalent to a constant energy shift, which is irrelevant for the weight as well as the physics). We can therefore assume $\boldsymbol{A}$ to be anti symmetric without loss of generality. We can therefore write

$$\sum_{ij} f_i f_j^\dagger A_{ij} = \sum_{ij} -f_j^\dagger f_i A_{ij} = -\sum_{ij} f_i^\dagger f_j A_{ji} = \sum_{ij} f_i^\dagger f_j A_{ij}. \tag{151}$$

Substituting this leads to

$$\begin{aligned}\langle\theta| \exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\Delta_x \boldsymbol{A}\left(\boldsymbol{f}^\dagger + \boldsymbol{f}\right)\right)|\psi\rangle &= \langle\theta|\psi\rangle - \frac{\Delta_x}{4}\langle\theta|\sum_{ij}\left(f_i^\dagger f_j^\dagger + 2f_i^\dagger f_j + f_i f_j A_{ij}\right)\boldsymbol{A}_{ij}|\psi\rangle + \mathcal{O}\left(\Delta_x^2\right) \\ &= \langle\theta|\psi\rangle - \frac{\Delta_x}{4}\langle\theta|\sum_{ij}\left(\bar{\theta}_i + \psi_i\right)\boldsymbol{A}_{ij}\left(\bar{\theta}_j + \psi_j\right)|\psi\rangle + \mathcal{O}\left(\Delta_x^2\right) \\ &= \exp\left(-\frac{1}{2}\left(\bar{\boldsymbol{\theta}} + \boldsymbol{\psi}\right)^{\mathrm{T}}\frac{\Delta_x \boldsymbol{A}}{2}\left(\bar{\boldsymbol{\theta}} + \boldsymbol{\psi}\right)\right) + \mathcal{O}\left(\Delta_x^2\right)\end{aligned} \tag{152}$$

So up to $\mathcal{O}\left(\Delta_x^2\right)$ the relation holds with $\boldsymbol{G}\left(\Delta_x\right) = \frac{\Delta_x \boldsymbol{A}}{2} + \mathcal{O}\left(\Delta_x^2\right)$ and $\eta\left(\Delta_x\right) = 1 + \mathcal{O}\left(\Delta_x^2\right)$. Next we assume the relation holds for the matrices $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ and look at the product of their exponentials:

$$
\begin{aligned}
&\langle\xi|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)|\psi\rangle\\
&\quad=\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\,\langle\xi|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)|\theta\rangle\langle\theta|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)|\psi\rangle\exp\left(-\bar{\theta}\theta\right)\\
&\quad=\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\,\langle\xi|\,\omega\left(\boldsymbol{A}_1,\bar{\xi}+\theta\right)|\theta\rangle\langle\theta|\,\omega\left(\boldsymbol{A_2},\bar{\theta}+\psi\right)|\psi\rangle\exp\left(-\bar{\theta}\theta\right)\\
&\quad=\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\;\omega\left(\boldsymbol{A}_1,\bar{\xi}+\theta\right)\omega\left(\boldsymbol{A_2},\bar{\theta}+\psi\right)\exp\left(-\bar{\theta}\theta+\bar{\xi}\theta+\bar{\theta}\psi\right)\\
&\quad=\eta_1\eta_2\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\;\exp\left(-\frac{1}{2}\left[\left(\bar{\xi}+\theta\right)\boldsymbol{G}_1\left(\bar{\xi}+\theta\right)+\left(\bar{\theta}+\psi\right)\boldsymbol{G}_2\left(\bar{\theta}+\psi\right)\right]\right)\\
&\qquad\qquad\qquad\qquad\qquad\times\exp\left(-\bar{\theta}\theta+\bar{\xi}\theta+\bar{\theta}\psi\right)\\
&\quad=\eta_1\eta_2\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\quad\exp\left(-\frac{1}{2}\begin{bmatrix}\bar{\xi} & \psi\end{bmatrix}\begin{bmatrix}\boldsymbol{G}_1 & 0\\ 0 & \boldsymbol{G}_2\end{bmatrix}\begin{bmatrix}\bar{\xi}\\ \psi\end{bmatrix}\right)\\
&\qquad\qquad\qquad\qquad\qquad\times\exp\left(-\frac{1}{2}\begin{bmatrix}\theta & \bar{\theta}\end{bmatrix}\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\begin{bmatrix}\theta\\ \bar{\theta}\end{bmatrix}\right)\\
&\qquad\qquad\qquad\qquad\qquad\times\exp\left(-\begin{bmatrix}\bar{\xi}\left(\boldsymbol{G}_1-\mathbb{I}\right) & \psi\left(\boldsymbol{G}_2+\mathbb{I}\right)\end{bmatrix}\begin{bmatrix}\theta\\ \bar{\theta}\end{bmatrix}\right)
\end{aligned}
\tag{153}
$$

We inserted a completeness relation for θ, used our assumption that the relation holds for $\boldsymbol{A}_{1,2}$ and rewrote the arguments of the exponentials in vector form. We note that this is of Gaussian form. Performing the integration leads to

$$
\begin{aligned}
&\langle\xi|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_1\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\boldsymbol{A}_2\left(\boldsymbol{f}^\dagger+\boldsymbol{f}\right)\right)|\psi\rangle\\
&\quad=(-1)^N\eta_1\eta_2\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right)\int \mathrm{d}\bar{\theta}\mathrm{d}\theta\quad\exp\left(-\frac{1}{2}\begin{bmatrix}\bar{\xi} & \psi\end{bmatrix}\begin{bmatrix}\boldsymbol{G}_1 & 0\\ 0 & \boldsymbol{G}_2\end{bmatrix}\begin{bmatrix}\bar{\xi}\\ \psi\end{bmatrix}\right)\\
&\qquad\qquad\qquad\times\exp\left(\frac{1}{2}\begin{bmatrix}\bar{\xi}\left(\boldsymbol{G}_1-\mathbb{I}\right) & \psi\left(\boldsymbol{G}_2+\mathbb{I}\right)\end{bmatrix}\underbrace{\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}^{-1}}_{:=\boldsymbol{F}^{-1}}\begin{bmatrix}\left(\boldsymbol{G}_1+\mathbb{I}\right)\bar{\xi}\\ \left(\boldsymbol{G}_2-\mathbb{I}\right)\psi\end{bmatrix}\right)
\end{aligned}
\tag{154}
$$

The alternating sign factor at the beginning emerges because we pulled out an overall sign from the pfaffian. The signs of the identities on the right vector have changed because we absorbed the sign from the matrix and then used the skew symmetry of $\boldsymbol{G}_{1,2}$. The matrix can be inverted by

$$
\boldsymbol{F}^{-1}=\begin{bmatrix}\boldsymbol{G}_2\left(\mathbb{I}+\boldsymbol{G}_1\boldsymbol{G}_2\right)^{-1} & \left(\mathbb{I}+\boldsymbol{G}_2\boldsymbol{G}_1^{-1}\right)\\ -\left(\mathbb{I}+\boldsymbol{G}_1\boldsymbol{G}_2\right)^{-1} & \boldsymbol{G}_1\left(\mathbb{I}+\boldsymbol{G}_2\boldsymbol{G}_1^{-1}\right)\end{bmatrix}.
\tag{155}
$$

With this, the previous expression can be rewritten as

$$\langle\xi|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\boldsymbol{A}_1\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\boldsymbol{A}_2\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\right)|\psi\rangle$$
$$= (-1)^N\eta_1\eta_2\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right)$$
$$\times\exp\left(-\frac{1}{2}\begin{bmatrix}\bar{\xi} & \psi\end{bmatrix}\begin{bmatrix}\boldsymbol{G}_1-(\boldsymbol{G}_1-\mathbb{I})\,\boldsymbol{G}_2\,(\mathbb{I}+\boldsymbol{G}_1\boldsymbol{G}_2)^{-1} & -(\boldsymbol{G}_1-\mathbb{I})\,(\mathbb{I}+\boldsymbol{G}_2\boldsymbol{G}_1)^{-1}\\ (\boldsymbol{G}_2+\mathbb{I})\,(\mathbb{I}+\boldsymbol{G}_1\boldsymbol{G}_2)^{-1}\,(\boldsymbol{G}_1+\mathbb{I}) & \boldsymbol{G}_2-(\boldsymbol{G}_2+\mathbb{I})\,\boldsymbol{G}_1\,(\mathbb{I}+\boldsymbol{G}_2\boldsymbol{G}_1)^{-1}\,(\boldsymbol{G}_2-\mathbb{I})\end{bmatrix}\begin{bmatrix}\bar{\xi}\\ \psi\end{bmatrix}\right) \tag{156}$$

By using the identity

$$(\mathbb{I}+\boldsymbol{G}_1\boldsymbol{G}_2)^{-1}=\frac{1}{2}\left(\mathbb{I}+\boldsymbol{K}_2\,(\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2)^{-1}\,(\mathbb{I}+\boldsymbol{K}_1)\right) \tag{157}$$

with some tedious calculations one can show that the expression reduces to

$$\langle\xi|\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\boldsymbol{A}_1\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\right)\exp\left(-\frac{1}{4}\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\boldsymbol{A}_2\left(\boldsymbol{f}^{\dagger}+\boldsymbol{f}\right)\right)|\psi\rangle$$
$$= (-1)^N\eta_1\eta_2\mathrm{Pf}\left(\begin{bmatrix}-\boldsymbol{G}_1 & \mathbb{I}\\ -\mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right)$$
$$\times\exp\left(-\frac{1}{2}\left(\bar{\xi}+\psi\right)\frac{\mathbb{I}-\boldsymbol{K}_1\boldsymbol{K}_2}{\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2}\left(\bar{\xi}+\psi\right)\right)\exp\left(\bar{\xi}\psi\right)$$
$$= \langle\xi|\,\eta_1\eta_2\mathrm{Pf}\left(\begin{bmatrix}-\boldsymbol{G}_1 & \mathbb{I}\\ -\mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right)\exp\left(-\frac{1}{2}\left(\bar{\xi}+\psi\right)\frac{\mathbb{I}-\boldsymbol{K}_1\boldsymbol{K}_2}{\mathbb{I}+\boldsymbol{K}_1\boldsymbol{K}_2}\left(\bar{\xi}+\psi\right)\right)|\psi\rangle \tag{158}$$

Here we notice that this expression is also of Gaussian form and we can identify

$$\boldsymbol{K}_3=\boldsymbol{K}_1\boldsymbol{K}_2 \qquad \text{and} \quad \eta_3=(-1)^N\eta_1\eta_2\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}_1 & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}_2\end{bmatrix}\right) \tag{159}$$

So we know the product of two expressions for which the relation we want to prove holds will again be an expression for which the relation holds. We also now how a product of such expressions combines. And lastly we know that for small exponents the relation holds. What we could do now is look at a long string of exponentials with small arguments. This is equivlaent of solving a differential equation. We know

$$\boldsymbol{K}(x+\Delta_x)=\boldsymbol{K}(x)\,\boldsymbol{K}(\Delta_x) \tag{160}$$

From eq.(149) we can show

$$\boldsymbol{K}(\Delta_x)=\frac{\mathbb{I}-\frac{\Delta_x\boldsymbol{A}}{2}}{\mathbb{I}+\frac{\Delta_x\boldsymbol{A}}{2}}=\mathbb{I}-\Delta_x\boldsymbol{A}+\mathcal{O}\left(\Delta_x^2\right) \tag{161}$$

Therefore eq.(160) can be reformulated as

$$\boldsymbol{K}(x+\Delta_x)=\boldsymbol{K}(x)\left(\mathbb{I}-\Delta_x\boldsymbol{A}\right), \tag{162}$$

from which we find the differential equation

$$\frac{\mathrm{d}\boldsymbol{K}}{\mathrm{d}x}=-\boldsymbol{A}\boldsymbol{K} \tag{163}$$

with the initial condition $\boldsymbol{K}(0)=1$. This is solved by

$$\boldsymbol{K}(1)=\boldsymbol{K}=e^{-\boldsymbol{A}}. \tag{164}$$

From this one easily finds

$$\boldsymbol{G}=\tanh\left(\frac{\boldsymbol{A}}{2}\right) \tag{165}$$

For η we can continue similarly:

$$\eta(x+\Delta_x)=(-1)^N\eta(x)\eta(\Delta_x)\,\mathrm{Pf}\left(\begin{bmatrix}\boldsymbol{G}(\Delta_x) & -\mathbb{I}\\ \mathbb{I} & \boldsymbol{G}(x)\end{bmatrix}\right)=(-1)^N\eta(x)\,\mathrm{Pf}\left(\begin{bmatrix}\Delta_x\frac{\boldsymbol{A}}{2} & -\mathbb{I}\\ \mathbb{I} & \tanh\left(x\frac{\boldsymbol{A}}{2}\right)\end{bmatrix}\right) \tag{166}$$

Here one can also find a differential equation:

$$\frac{\mathrm{d}\eta}{\mathrm{d}x}=\frac{1}{4}\mathrm{Tr}\left\{\boldsymbol{A}\tanh\left(x\frac{\boldsymbol{A}}{2}\right)\right\}\eta(x), \tag{167}$$

with $\eta(0)=1$. The solution of this is given by

$$\eta(1)=\eta=(-1)^N\,\mathrm{Pf}\left(\begin{bmatrix}\sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right) & -\mathbb{I}\\ \mathbb{I} & \sqrt{2}\sinh\left(\frac{\boldsymbol{A}}{4}\right)\end{bmatrix}\right) \tag{168}$$